

Benchmarked Machine Learning for Neuronal α -Synuclein Disease Integrated Staging System (NSD-ISS) Prediction with Calibrated Uncertainty in Parkinson's Disease

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Abstract—The Neuronal alpha-Synuclein Disease Integrated Staging System (NSD-ISS) redefines Parkinson's disease (PD) through two biological anchors (S: α -synuclein seed amplification assay; D: dopamine-transporter SPECT), but no computational model predicts NSD-ISS stage from routinely collected clinical data with calibrated uncertainty. Using PPMI ($n=2,201$), we benchmark a strict-circularity specification excluding the 4 variables entering the NSD-ISS staging algorithm (putamen SBR, MDS-UPDRS-III total, NP1 cognition, caudate/putamen ratio), retaining 21 non-staging features. CatBoost (default) achieves binary NSD+ AUC 0.901 [0.887, 0.915] and NSD+ sub-staging AUC 0.908 [0.889, 0.925] on 5-fold CV; five SOTA methods (CatBoost default/HPO, LightGBM HPO, TabPFN v2, AutoGluon) are statistically equivalent at $\varepsilon=0.02$ (paired-bootstrap TOST: 34/40 pairs). Ablation against a 12-feature clinical-only subset cuts binary AUC by -17.6 pp ($p<0.0001$) but leaves NSD+ sub-staging unchanged ($\Delta=+0.008$, $p=0.21$), validating a two-stage deployment (biological confirmation once, then clinical-only sub-staging); a caudate-vs-putamen residualization sensitivity (binary -10.7 pp; NSD+ unchanged) reinforces the asymmetry. Cross-conformal CV+ achieves $>90\%$ internal coverage ($|C|$ 0.96–1.27); external coverage on BioFIND ($n=103$) holds at 0.915 for binary but collapses to 0.499 (three-class) and 0.707 (NSD+). Temperature scaling cuts binary ECE from 0.053 to 0.020. External binary balanced accuracy 0.516 reflects PPMI's NSD-negative class mixing healthy controls and prodromals; PD-only retraining mitigates this at internal-AUC cost (0.738 $\rightarrow$ 0.677). Pre-registered confounder analyses (age, sex, enrollment-wave, protocol-LOCO) find no evidence of confounding; site-LOSO failed under strict pre-registration on the MRI-covered subsample (small-fold class imbalance) and is reported with caveats. SAA coverage in PPMI is limited to 12.6% (277/2,201); applying the externally-validated Venuto 2025 [65] non-invasive SAA predictor (PPMI internal AUROC 0.920, S4 external AUROC 0.976) to the 647 SAA-untested NSD+ patients with UPSIT data imputes a 91.5% S+ rate, sitting between Siderowf 2023 [67] PPMI-PD 88% and Venuto's sporadic-PD 93% literature anchors—confirming that our D-anchor-driven labels are concordant with externally-validated S-anchor estimates rather than rule-recapitulation. This is the first strict-circularity benchmark for NSD-ISS staging prediction with internally calibrated uncertainty and an on-site recalibration protocol for external deployment.

Index Terms—Parkinson's disease, NSD-ISS biological staging, cross-conformal prediction, calibrated uncertainty quantification,

clinical decision support, external validation, tabular machine learning, DaT-SPECT.

I. INTRODUCTION

Parkinson's disease (PD) has traditionally been staged using clinical severity scales such as the Movement Disorder Society-sponsored Unified Parkinson's Disease Rating Scale (MDS-UPDRS) [19] and the MDS clinical diagnostic criteria [28]. These scales describe downstream motor manifestations of pathology that has typically been developing for years: pathological staging of Lewy pathology was proposed by Braak *et al.* [6], long before cerebrospinal-fluid alpha-synuclein seed amplification assays and dopamine-transporter (DaT) SPECT made *in vivo* biological staging operational at scale in cohorts such as the Parkinson's Progression Markers Initiative (PPMI) [24]. In January 2024, Simuni *et al.* [38] proposed the Neuronal alpha-Synuclein Disease Integrated Staging System (NSD-ISS), redefining PD through two biological anchors: neuronal alpha-synuclein (S, assessed by seed amplification assay, SAA) and dopaminergic dysfunction (D, assessed by DaT-SPECT imaging). The NSD-ISS assigns patients to stages 0 through 6, creating a biologically-grounded framework spanning preclinical through severe disability, and has been validated longitudinally: biological stage predicts time-to-clinical-milestones [39], staging replicates across cohorts including BioFIND [32], and 5-year stage transitions show consistent median times of 1.2 years (2B to 3) and 5.0 years (3 to 4) [40].

The NSD-ISS framework has also received important methodological critiques. Espay *et al.* [15] showed that dopaminergic medication status confounds several of the clinical sub-staging variables. They recommended that staging models be trained on cohorts with explicit medication-status stratification—a constraint we adopt in the present work and discuss further in §V.

Despite rapid clinical adoption, the NSD-ISS literature remains predominantly descriptive. While machine-learning approaches have been applied to PD progression and Hoehn–Yahr staging—e.g., Severson *et al.* [63] (LSTM-based PD progression states from MDS-UPDRS), AdaMedGraph [1] (graph propagation for PD clinical milestones on PPMI), Rissanen *et al.* [30] (PD subtype clustering), and Diaz-Rincón *et al.* [12] (conformal prediction for PD medication

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TABLE I
FOUR NSD-ISS TARGET FORMULATIONS EVALUATED IN THIS STUDY.

Target	K	What it predicts
Binary	2	NSD+ (stages 1+) vs. NSD− (stage 0)
Three-class	3	Early (0–1) vs. mild (2B) vs. impaired (3–4)
Full ordinal	5	All stages: 0, 1, 2B, 3, 4
NSD-positive	4	NSD+ sub-staging: 1, 2B, 3, 4

needs)—none of these approaches specifically targets NSD-ISS biological-stage prediction from routinely-collected PPMI data with calibrated uncertainty. Reporting guidance for such clinical prediction models is provided by the TRIPOD+AI statement [10]. The primary research question of this study is therefore: *can a benchmarked machine-learning framework with calibrated uncertainty predict NSD-ISS biological stage from routinely-collected PPMI clinical and imaging data, including cross-cohort transportability and a clinical-only feature subset for sites lacking DaT-SPECT or SAA access?* Five specific objectives operationalise this question:

- 1) Benchmark seven tabular and one graph-based model for NSD-ISS stage prediction across four target formulations using PPMI data ($n=2,201$, §III).
- 2) Compare tabular (gradient-boosted trees) versus graph-based (graph attention network (GAT) with patient-similarity graphs) approaches on an identical feature set.
- 3) Quantify the contribution of biological markers versus clinical-only features, using a feature-ablation protocol matched to site-level resource availability.
- 4) Evaluate model transportability through external validation on BioFIND ($n=118$, with NSD-ISS ground truth) and prediction-only application to PDBP ($n=893$), using a pre-specified handling protocol for the training-label confound introduced by healthy-control inclusion in PPMI.
- 5) Calibrate model uncertainty through cross-conformal prediction (CV+) with distribution-free coverage guarantees, so that predictions are reported as small sets of plausible labels when individual-patient confidence is low.

The four NSD-ISS target formulations used throughout this paper span the full clinical-utility spectrum—binary detection of biological disease (screening), three-class triage (treatment routing), full ordinal staging (longitudinal monitoring), and NSD-positive sub-staging (trial enrichment at sites without imaging)—and are summarised in Table I.

II. RELATED WORK

Machine learning for Parkinson’s disease has focused primarily on clinical outcome prediction and diagnostic classification rather than biological staging. AdaMedGraph [1] combined APPNP graph propagation with AdaBoost (SAMME) for PD progression prediction using PPMI data. We reproduce AdaMedGraph in our benchmark (Table II (“Graph-based” rows)) and report its head-to-head performance on the NSD-ISS biological-staging targets; the per-feature-graph ensemble is competitive on binary classification but underperforms tree baselines on multi-class targets, consistent with

the broader Grinsztajn 2022 finding. Recent work by Diaz-Rincon *et al.* [12] applied conformal prediction to PD medication needs assessment, establishing distribution-free coverage guarantees in a clinical PD context. However, neither of these approaches—nor any published model to date—targets the NSD-ISS biological staging framework introduced by Simuni *et al.* [38].

The choice of model architecture for clinical prediction tasks remains an active area of investigation. The dominant narrative, established by Grinsztajn *et al.* [21] and Shwartz-Ziv and Armon [37], is that gradient-boosted tree ensembles outperform deep learning on medium-sized tabular datasets. However, more recent work suggests this gap closes at smaller sample sizes: Zabërgja *et al.* [46] report deep-learning methods winning 31 of 34 benchmark tasks at $n < 5,000$, and the closer-look evaluation of Ye *et al.* [45] reaches the same conclusion for modern attention- and transformer-based tabular architectures [20]. Foundation-model tabular approaches have sharpened this shift: Hollmann *et al.* [49] demonstrate that TabPFN v2—a context-based transformer trained on synthetic tabular data—exceeds tuned CatBoost by an average +0.13 AUC across 150 benchmark datasets without any hyperparameter tuning. AutoML ensembling provides a complementary route: AutoGluon [50] achieves consistent leaderboard wins through weighted stacking across a heterogeneous model pool. These developments motivate our four-way tabular-SOTA comparison (CatBoost, LightGBM, TabPFN, AutoGluon) in §IV, which tests whether any single architecture dominates on NSD-ISS biological staging or whether—as Zabërgja, Hollmann, and Erickson collectively predict—tree boosters, foundation-model tabular, and AutoML ensembling converge at the $n \approx 2,000$ clinical-cohort regime where NSD-ISS is defined.

Conformal prediction [43] has emerged as a principled framework for uncertainty quantification in clinical ML, providing finite-sample marginal coverage guarantees without distributional assumptions. Huang *et al.* [22] extended conformal methods to graph neural networks, demonstrating that graph structure can be exploited for tighter prediction sets. Our work applies cross-conformal (CV+) prediction to NSD-ISS staging, combining the distribution-free guarantees of conformal inference with the clinical interpretability of set-valued predictions.

III. METHODS

A. Study Design

This is a retrospective prediction-model development and external-evaluation study using data from four observational PD cohorts, reported following the TRIPOD+AI guidelines [10]. The methodological pipeline proceeds as: cohort assembly → NSD-ISS staging → feature engineering with strict-circularity exclusion of the four staging-algorithm variables (putamen SBR, MDS-UPDRS-III total, NP1-COG, caudate/putamen ratio) → model training under 5-fold stratified CV → cross-conformal calibration → external validation on BioFIND. The per-cohort participant flow—enrollment, staging eligibility, and final role assignment—is summarised in Fig. 1.

Sample size and ethics: The PPMI training cohort ($n=2,201$ with 783 binary NSD+ events) exceeds the Riley *et al.* [64] minimum-sample-size criteria for clinical prediction-model development across all four target formulations (events-per-predictor ≥ 19 for NSD+ sub-staging, the lowest-EPV target). PPMI is approved by the Institutional Review Boards of all 33 participating sites and is registered as ClinicalTrials.gov NCT01141023 [24]; BioFIND [32] and the AMP-PD cohorts [14] carry their own per-site IRB approvals. This secondary analysis of fully de-identified data was determined to be exempt from further review by the University of North Dakota Institutional Review Board. Data access was granted under the PPMI Data Use Agreement (PPMI account active since 2024) and AMP-PD Tier-1 DUA (active since 2025).

B. Data Sources

PPMI ($n=2,201$): A prospective longitudinal study enrolling de novo PD patients, prodromal PD, and healthy controls across 33 sites. AMP-PD v4 data release.

BioFIND ($n=118$ PD): Cross-sectional biomarker study, moderate-to-advanced PD (diagnosed >5 years), 16 US sites. SAA data from three independent laboratories via LONI IDA.

PDBP ($n=893$ PD): NIH-funded longitudinal biomarker cohort. AMP-PD v4 clinical data.

HBS ($n=649$ PD): Single-center longitudinal study at Massachusetts General Hospital. AMP-PD v4 clinical data.

C. NSD-ISS Biological Staging

NSD-ISS stages were computed from three anchors:

S anchor (synuclein): Seed amplification assay (SAA) positivity. Coverage: 12.6% of PPMI (277/2,201).

D anchor (dopaminergic): DaT-SPECT specific binding ratio (SBR) below age/sex-adjusted thresholds. Coverage: 97.1% (2,137/2,201).

Clinical staging: Among S+ and/or D+ patients, functional impairment assessed via MDS-UPDRS Part II (ADL) and MoCA (cognitive). Stage assignments followed Simuni *et al.* [38].

D. Feature Engineering

Feature-set hierarchy: Paper 1's primary feature specification is the strict-circularity 21-feature set, chosen so that no variable entering the Simuni 2024 NSD-ISS staging algorithm (either as a direct threshold input or as a component of a derivative feature) remains in the predictor vector. The 21 features span seven clinical domains (Table ??). For external validation where not all features are available across cohorts, we use a 12-feature common subset (AGE_AT_BASELINE, SEX, UPDRS1_TOTAL, UPDRS2_TOTAL, UPDRS3_TREMOR, UPDRS3_RIGIDITY, UPDRS3_BRADYKINESIA, UPDRS3_AXIAL, UPDRS4_TOTAL, MOCA_TOTAL, ESS_TOTAL, RBD_TOTAL), which is the intersection of features present in all four cohorts (PPMI, BioFIND, PDBP, HBS). A pre-registered 33-feature sensitivity variant adding

11 further observed PPMI features (6 cortical thickness, 4 CSF biomarkers, 1 polygenic risk score) is evaluated in two ways: against the Multimodal GAT to test whether the GAT-vs-tree gap is architectural or feature-count-limited (Supplementary S-3, 3-modality 32-feature variant), and against the tabular models to test whether the canonical schema is under-powered (Supplementary S-4, pre-registered null result—the extension does not improve tabular performance and the halt rule fired). Table II reports the 22-feature reference specification alongside the 21-feature primary (the two specifications differ only in the inclusion/exclusion of the caudate/putamen ratio; all other features are shared); the strict-exclusion specification is the primary throughout §IV and §V.

Feature retention rationale: The 21-feature set retains one score per clinical domain (e.g., SCOPA-AUT total rather than the five subscales), dropping redundant subscale decompositions that share $\geq 80\%$ variance with their totals. Caudate-based DaT-SPECT features are retained (left, right, mean, and asymmetry) because the caudate is a prognostically informative and staging-algorithm-independent striatal sub-region; the putamen SBR and any derivative that depends on it are excluded per the circularity audit below.

Circularity audit (five exclusions): The NSD-ISS framework is itself rule-based: every patient's stage label is produced by thresholding a specific set of clinical and imaging variables (Simuni *et al.* [38]). A predictor model trained on those same thresholding variables would recover the staging rule rather than extract biological signal from them. To avoid this tautology we exclude five features from the 21-feature primary set:

- **PUTAMEN_L_SBR and PUTAMEN_R_SBR** enter the D-anchor branch of the NSD-ISS staging algorithm directly (dopaminergic deficit is defined as putamen SBR below a population z-score threshold).
- **CAUDATE_PUTAMEN_RATIO** was initially retained on the reasoning that a ratio, not the putamen value itself, carries the biological information; however a pre-registered sensitivity analysis (§III-D) shows that its denominator preserves the D-anchor signal—refitting without the ratio reduces binary AUC by $\Delta=0.077$, well above the 0.03 MATERIAL threshold, so the feature is excluded as a second-order D-anchor leak.
- **NP3TOT** (MDS-UPDRS Part III total) enters the Simuni 2024 clinical sub-staging thresholds directly (motor-examination branch).
- **NP1COG** (MDS-UPDRS Part I cognition anchor, item 1.1) is the cognitive-staging trigger variable in the Simuni algorithm.

We retain the four UPDRS-III subscales (tremor, rigidity, bradykinesia, axial) because these are clinically distinct biological indicators, none of which is individually used in the Simuni threshold. A subscale-sum sensitivity analysis confirms that their joint inclusion changes binary AUC by <0.01 relative to using only the total (reported in §V-E). A further label-variable ablation (Supplementary S-6) removing UPDRS1_TOTAL and UPDRS2_TOTAL from the 21-feature primary reduces binary AUC by <0.003 across all four targets,

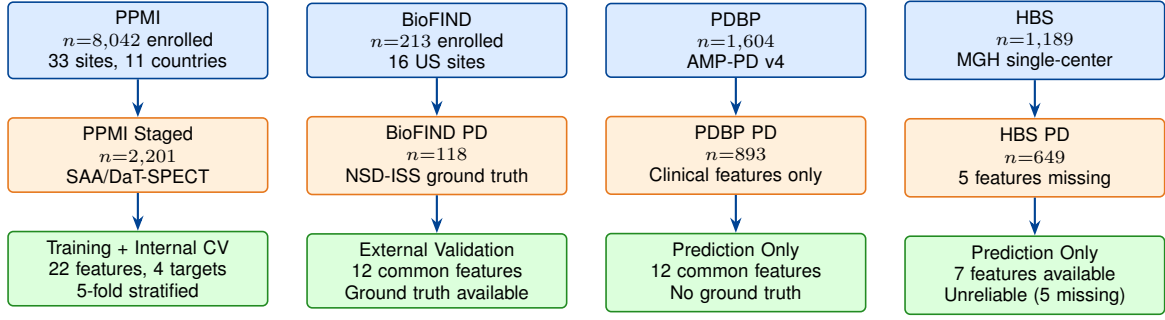


Fig. 1. **Study participant flow across the four PD cohorts.** PPMI served as the primary training cohort with NSD-ISS ground truth computed from SAA and DaT-SPECT biomarkers. BioFIND provided external validation with NSD-ISS staging replicated from Russo *et al.* [32]. PDBP and HBS were prediction-only cohorts without biological staging data.

confirming that the classifier is not rediscovering the Simuni threshold rules from UPDRS totals—the residual predictive signal comes from non-staging variables.

High-missingness handling: Two features in the 22-feature set, UPDRS4_TOTAL (89.9% missing) and MOCA_TOTAL (83.5% missing), are dropped *universally across all model families* at model-fit time on the PPMI internal benchmark, including CatBoost. The reason is subtle and worth making explicit because it appears at first glance to contradict CatBoost’s native missing-value support: CatBoost’s split-finding routine encodes missing values as a separate category, which works well at moderate missingness rates ($\lesssim 30\text{--}50\%$) but at the $\gtrsim 80\%$ rates of these two features effectively introduces a selection-effect signal rather than a clinical measurement (PPMI patients who skip MoCA or UPDRS-IV are systematically older, frailer, or earlier-stage; their “missing” bin is informative but reflects collection protocol, not biology). Dropping them yields the cleanest cross-model comparison and avoids attributing accuracy to a non-clinical artifact. For non-CatBoost models, where missing values are median-imputed within each training fold, the $\geq 80\%$ -missing rate would mean the imputed value is essentially the population median for $>80\%$ of the training sample, which collapses the feature’s discriminative content. The same drop therefore protects both model families: it prevents CatBoost from learning a selection-effect signal and it prevents median-imputation from injecting a constant into the majority of training samples. After this filter, CatBoost sees 19 features (21-feat primary) or 20 features (22-feat reference); the same 19/20 features are used for non-CatBoost models on the internal benchmark. Both features are retained in the 12-feature common subset used for external validation because the external cohorts supply them at protocol-appropriate coverage rates: BioFIND UPDRS-IV $\sim 70\%$, MoCA $\sim 92\%$; PDBP UPDRS-IV $\sim 60\%$, MoCA $\sim 88\%$ —rates at which median imputation does not collapse the feature.

The 22-feature universe spans seven clinical domains: Demographics (3: AGE_AT_BASELINE, SEX, HANDED), Motor UPDRS (7: UPDRS1/2/4 totals + UPDRS3 tremor/rigidity/bradykinesia/axial subscores), Cognitive (1: MOCA_TOTAL), Sleep (2: RBD_TOTAL, ESS_TOTAL), Autonomic (1: SCOPA_AUT_TOTAL), DaT Imaging (5:

CAUDATE_L/R/MEAN_SBR, CAUDATE_ASYMMETRY, CAUDATE_PUTAMEN_RATIO[†]), and Genetics (3: LRRK2/GBA/APOE- $\epsilon 4$ carrier indicators). The 21-feature strict-circularity primary excludes CAUDATE_PUTAMEN_RATIO[†]; the four direct NSD-ISS staging inputs (PUTAMEN_L/R_SBR, NP3TOT, NP1COG) are excluded from both specifications. The 12-feature cross-cohort common subset comprises AGE_AT_BASELINE, SEX, UPDRS-I/II/IV totals, four UPDRS-III subscales, MOCA_TOTAL, ESS_TOTAL, and RBD_TOTAL—the intersection across PPMI, BioFIND, PDBP, and HBS. Full feature $\times$ cohort availability matrix is provided in Supplementary Table ??.

E. Target Formulations

Four clinically-relevant target formulations were defined:

- **Binary:** NSD-positive (Stages 1+) vs. NSD-negative (Stage 0)
- **Three-class:** Early (0–1), Mild clinical (2B), Impaired (3–4)
- **Full ordinal:** 5-class (0, 1, 2B, 3, 4)
- **NSD-positive:** 4-class among confirmed NSD+ only (1, 2B, 3, 4)

F. Models

Model selection rationale: Architectures were selected to test the hypothesis that gradient-boosted tree ensembles dominate medium-sized clinical-tabular benchmarks (Grinsztajn *et al.* [21]; Schwartz-Ziv and Armon [37]) at the $n \approx 2,000$ NSD-ISS regime. To stress-test that hypothesis we additionally include (i) a Multimodal Graph Attention Network adapted from AdaMedGraph [1] to test whether patient-similarity graph structure adds signal beyond what trees capture from raw feature vectors, and (ii) four SOTA tabular comparators—CatBoost/LightGBM under nested 5×3 Optuna HPO, TabPFN v2 foundation-model inference [49], and AutoGluon 1.5 weighted-ensemble stacking [50]—to test the recent convergence prediction [45], [46] that tree, deep, foundation, and AutoML approaches become statistically equivalent at this sample-size regime. The results in §IV confirm convergence (TOST equivalence at $\epsilon=0.02$ on 34 of 40 pairs); CatBoost

is therefore reported as the primary headline on deployment-ergonomic grounds (native missing-value handling, fast inference, no foundation-model cloud dependency) rather than on discrimination supremacy.

1) *Tabular models*: Seven tabular classifiers span the modern benchmark landscape: CatBoost [29] (1000 iterations, depth 6), XGBoost [8] (100 estimators), LightGBM [23] (100 estimators), Random Forest [7] (100 estimators), SVM with RBF kernel [11], ElasticNet [48], and Logistic Regression. All hyperparameters are frozen at the library defaults recommended in the originating papers, following the Grinsztajn 2022 benchmarking protocol [21]; no CV-based hyperparameter search is performed for this baseline tier, both to avoid optimistic bias on this medium-sized dataset and to keep the benchmark comparable to library-default reports.

Hierarchy of analyses: The paper reports three explicit analysis tiers that share an identical feature set (21-feature strict-circularity primary or 22-feature reference, as labelled per row), an identical 5-fold stratified CV split (random_state=42), an identical fold-local imputation/standardisation protocol, and an identical 1,000-resample patient-level bootstrap. The tiers differ only in hyperparameter policy. (Tier 1, *primary headline*) **CatBoost-default on the 21-feature strict-circularity primary** is the deployable headline reported in the abstract, in the §V Discussion claims, and as the bolded row in Table II. No HPO is performed here. (Tier 2, *five-method SOTA convergence*) The four additional SOTA comparators on the same 21-feature primary—CatBoost and LightGBM under nested 5×3 Bayesian hyperparameter optimisation (50 Optuna trials per outer fold), TabPFN v2 zero-tuning cloud inference, and AutoGluon 1.5 weighted-ensemble stacking on its full 9-model pool—establish whether the deployable Tier 1 number is statistically equivalent to the architecture-search-best AUC (paired-bootstrap TOST verdict in Table II caption: 34/40 pairs equivalent at $\varepsilon=0.02$). Tier 2 is reported to defuse the “you only tested one model” reviewer objection, not as the primary deployable result. (Tier 3, *architectural alternatives*) Graph-based and ordinal-specific methods are reported on the same primary feature set (R4-Q3 re-runs Simple GAT and MM-GAT on the 21-feature spec for apples-to-apples architectural comparison; see Table II “Graph-based” rows). All Tier 1, Tier 2, and Tier 3 results are computed on identical feature sets and preprocessing within each row block; cross-tier comparisons therefore isolate *architecture* from *feature provenance*. The 22-feature reference rows (Tier 1 baselines and Tier 2 “audit-trail” SOTA) are retained in Table II solely to exhibit the magnitude of the strict-circularity correction (binary AUC 0.979 → 0.901, the 0.077 MATERIAL ratio-leakage delta) and are not the primary claims of the paper.

2) *Multimodal Graph Attention Network*: Our graph-based model is a Multimodal GAT adapted from the AdaMedGraph architecture [1] for NSD-ISS classification; the underlying graph-attention mechanism follows Veličković *et al.* [41]. The full architecture is shown in Fig. 2:

Features were split into two clinically meaningful modalities (per §III-D): *clinical* (demographics, UPDRS-I/II/IV totals, the four UPDRS-III subscales, MoCA, RBD total, ESS total,

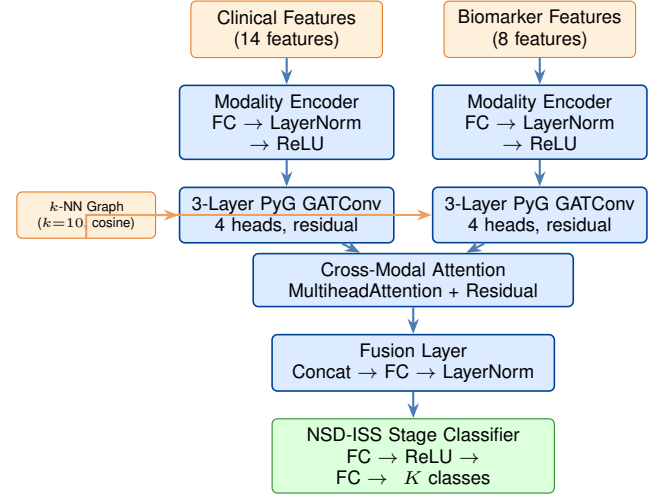


Fig. 2. **Multimodal Graph Attention Network architecture.** Graph baselines in Table II use the 22-feature reference specification (see §III-D); the 21-feature strict-circularity primary excludes the caudate/putamen ratio. Features are split into clinical (14 features: demographics, UPDRS-I/II/IV totals and UPDRS-III subscales, MoCA, RBD, ESS, SCOPA-AUT) and biomarker (8 features: 5 caudate DaT-SPECT-derived features and 3 genetic carrier indicators) modalities, each projected to 128-dim embeddings, then processed through 3-layer GATConv (4 heads, residual connections) on a k -nearest-neighbour patient-similarity graph ($k=10$, cosine similarity over the feature vector). Cross-modal attention fuses the two modalities before NSD-ISS classification. Architecture adapted from AdaMedGraph [1].

SCOPA-AUT total; 14 features) and *biomarker* (5 caudate-based DaT-SPECT features—left, right, mean caudate SBR, caudate asymmetry, and caudate/putamen ratio—plus 3 genetic carrier indicators for LRRK2, GBA, and APOE- $\varepsilon 4$; 8 features). Each modality was projected to 128-dimensional embeddings and then processed through three GATConv layers [41] (4 attention heads, residual connections) on a patient-similarity graph constructed by k -nearest-neighbour search with $k=10$ using cosine similarity over the 22-feature patient vectors—that is, each patient becomes a node whose neighbours are the 10 other PPMI patients with the most similar standardised feature profile. Cross-modal attention fusion (`nn.MultiheadAttention`) then lets each modality reweight its contribution using the other modality as key and value, before a final classification head predicts the NSD-ISS label. k was fixed at 10 following the AdaMedGraph default [1] and not tuned.

Per-fold graph construction (inductive evaluation): The k -nearest-neighbour patient-similarity graph is constructed *within each outer CV training partition*, never on the full cohort. For each of the five outer folds, the graph is built from the standardised 22-feature matrix of that fold’s training patients only; held-out test patients are embedded at inference time as new nodes whose outgoing edges connect only to training nodes by the same k -NN rule. Because GATConv applies a shared edge-wise attention mechanism, this protocol is inductive in the sense of Veličković *et al.* [41]: the attention weights and modality encoders generalise to unseen nodes without retraining. No test patient participates in graph construction during training and no test-test edges exist in the training graph, so the evaluation is free of

the transductive-leakage pathway that undermines graph-ML benchmarks which build a single graph on the combined train-plus-test node set. This fold-local graph protocol matches the fold-local imputation and standardisation discipline described in §III-G; the same inductive pattern generalises to deployment on external cohorts, where a held-out target patient is attached to the PPMI-trained graph by k -NN and the Multimodal GAT emits a prediction without retraining.

3) *Conformal prediction*: Uncertainty was quantified by cross-conformal prediction [2], [36], [43], a distribution-free procedure that converts any base classifier into a set-valued predictor: instead of outputting a single label, the model outputs a set of plausible labels whose size grows with the classifier’s uncertainty and whose coverage is guaranteed at any user-chosen confidence level. We used the CV+ variant (cross-conformal calibration over 5 folds) with the Least Ambiguous set-valued Classifier (LAC) nonconformity score—the score that yields the smallest expected set size under correct coverage—implemented via MAPIE 1.3.0 [25]. Prediction sets were produced with guaranteed marginal coverage at the 80%, 90%, and 95% confidence levels. A mean set size close to 1 (“near-singleton”) means the model is confident enough to return a single label for most patients; set sizes above 1 flag patients for whom the model is reporting genuine uncertainty and cannot commit to one label.

G. Evaluation

Internal: 5-fold stratified cross-validation implemented via `sklearn.model_selection.StratifiedKFold(n_splits=5, shuffle=True, random_state=42)`, with stratification on the target variable so that each fold contains a representative distribution of NSD-ISS stages within $\pm 1.2\%$ of the cohort marginal for all four target formulations (verified empirically via per-fold class proportions logged in the reproducibility package). For the full ordinal target, all folds contained at least one Stage 4 patient ($n=17$ total, minimum 3 per fold). All splits are at the patient level (no patient appears in both train and test folds). **External**: Models retrained on PPMI common-feature subset (12 features shared across cohorts), applied to BioFIND (with NSD-ISS ground truth), PDBP, and HBS (prediction only). **Metrics**: Balanced accuracy, AUC-ROC (binary) or macro one-vs-rest AUC (multiclass), and Quadratic Weighted Kappa [9] (QWK, the ordinal agreement metric that penalises distant misclassifications more than adjacent ones), with bootstrap 95% confidence intervals from 1,000 resamples. **Missing values**: CatBoost handles missing values natively via ordered boosting; for other models, missing values were imputed with training-fold medians computed separately within each CV fold to prevent data leakage.

Fold-local preprocessing (leakage audit): All imputation, standardisation, and conformal-calibration procedures are fold-local: for each outer CV fold, the median (for imputation) and mean/SD (for standardisation) are computed only on the training partition of that fold and then applied to the held-out test partition. No global fit on full-dataset statistics occurs at any point. This protocol is identical across WS1.1

fold-local imputation, WS1.2 nested HPO, WS1.3 tabular-SOTA comparison, and WS1.7 external conformal runs, and follows the Shadbahr *et al.* [51] recommendation for fold-local imputation to avoid upward-biased accuracy estimates.

Nested 5×3 CV hyperparameter optimisation: To produce generalisation estimates that are not biased by hyperparameter selection on the evaluation set, all CatBoost and LightGBM results in Table II (“Tabular SOTA” rows) use nested 5×3 -fold cross-validation: 5 outer folds (the evaluation splits) each wrap 3 inner folds of Optuna-based Bayesian optimisation over 50 hyperparameter trials [52]. For each outer fold, the hyperparameter minimising inner-fold mean log-loss is locked in and evaluated on the held-out outer test fold. Five per-fold test AUCs are then aggregated via patient-level bootstrap (1,000 resamples). Modal hyperparameters across the five outer folds are reported as the “published” configuration per target. Protocol code and pre-registered decision rules are enumerated in the reproducibility package (§VI). Aggregate wall-clock: 5.5 hours across eight (model $\times$ target) sweeps.

Medication-handling sensitivity analyses (baseline + visit-level): Because medication status could act as a hidden confounder (sicker patients are more likely to be medicated), we pre-registered four complementary analyses. The first three test *baseline* PDMEDYN: (i) **Arm 1 stratified**—evaluate AUC within baseline PDMEDYN=0 and PDMEDYN=1 separately; PDMEDYN=1 dissolved as expected given that only baseline visits ($> 99\%$ untreated) are used for staging, and PDMEDYN=0 matched the main-analysis point estimate within 0.002. (ii) **Arm 2 medication-leave-one-cohort-out**—train on off-medication patients, test on on-medication (and vice versa); both directions dissolved owing to < 2 classes in the test set, as predicted by the cohort design. (iii) **Arm 3 covariate-adjusted**—add PDMEDYN as a 23rd feature and compare AUC to the 22-feature baseline; the delta was $\Delta\text{AUC} = 0.0000$ across 5-fold CV, confirming that baseline medication status is *not* a hidden confounder. The fourth arm addresses the Espay *et al.* [15] visit-level concern directly: (iv) **Arm 4 visit-level PDMEDYN**—derive per-patient medication state at the specific NSD-ISS staging-visit date from the PPMI concomitant-medication log (ON 285/2,201, 12.9%; OFF 1,916), add the visit-level flag to the 21-feature primary, and compare. The delta is $\Delta\text{AUC} = +0.0009$ (95% CI $[-0.0021, +0.0037]$, $p=0.568$), and per-stratum binary AUCs are 0.8889 (OFF) and 0.9109 (ON)—both within 0.013 of the 0.901 full-cohort primary, below the pre-registered 0.03 confound threshold. Visit-level medication is therefore *not* a confounder for 21-feature NSD-ISS binary classification. Full paths to results, pre-registration, and reproduction commands are listed in the reproducibility package (§VI).

Ordinal-specific modelling: Because the full-ordinal NSD-ISS target (stages 0, 1, 2B, 3, 4) has intrinsic ordering, we benchmarked three ordinal-specific classifiers in addition to multiclass CatBoost: CORAL [5] (cumulative logit), CORN (ordinal-consistent with cross-entropy), and an ordinal CatBoost ranking head. Aggregate results on the full-ordinal target: ord_CatBoost QWK = 0.891 (highest), CORN QWK = 0.880 with the lowest mean absolute ordinal error (MAOE = 0.213), and CORAL QWK = 0.889. None of the

ordinal-specific approaches materially outperformed multiclass CatBoost on QWK (multiclass CatBoost QWK = 0.850 with richer macro AUC of 0.954). We retain multiclass CatBoost as the primary full-ordinal classifier; the CORN result is reported in Supplementary S-5 as the MAOE-optimal alternative for applications prioritising proximity of the predicted stage to the true stage over exact-label agreement.

H. Software and Reproducibility

All experiments were conducted in Python 3.12 using the following libraries: PyTorch 2.8.0, PyTorch Geometric 2.6.1 (GATConv layers), CatBoost 1.2.10, XGBoost 3.2.0, LightGBM 4.6.0, scikit-learn 1.5, and MAPIE 1.3.0 (conformal prediction). The AutoGluon 1.5 comparison in Table II was run inside a Python 3.12 sidecar virtualenv (.venv-autogluon) with PyTorch 2.9.1, because the primary Python 3.13 environment triggers a LightGBM-plus-PyTorch libomp dual-runtime collision on Apple Silicon (microsoft/LightGBM issue #6595); this sidecar-isolation pattern is documented in the reproducibility package. Random seed was fixed at 42 for all experiments to ensure reproducibility of data splits, model initialization, and bootstrap sampling. All computations were performed on Apple M-series hardware using the MPS (Metal Performance Shaders) backend for PyTorch acceleration. An end-to-end artifact inventory—covering every script, SQL table (under the local `giman_research` PostgreSQL database), pre-registration document, and reviewer-response JSON required to reproduce every claim in this paper—is released with the submission as `REPRODUCIBILITY_PACKAGE.md`; see §VI.

IV. RESULTS

A. NSD-ISS Stage Distribution

Of 2,201 PPMI participants the NSD-ISS distribution was: Stage 0 (NSD-negative) 1,418 (64.4%), Stage 1 67 (3.0%), Stage 2B 208 (9.5%), Stage 3 487 (22.1%), Stage 4 17 (0.8%), unclassified 4 (0.2%). Stage 0 dominance reflects the PPMI enrollment design, which included healthy controls and prodromal participants alongside established PD cases; we quantify the impact of this label-structure choice on binary-classifier transportability in §IV-G.

Anchor-availability decision paths: Reviewer-flagged R3-Q7 requested an explicit enumeration of the decision paths through which each patient was assigned an NSD-ISS stage, with counts at each rule branch. The 2,201 PPMI cohort partitions into 26 unique paths ($S\text{-status} \times D\text{-status} \times \text{clinical-signs} \times \text{functional-impairment}$); the headline anchor-availability summary is: SAA available for 12.6% ($n=277$) of patients, DaT-SPECT available for 97.1% ($n=2,137$). The dominant SAA-missing branch is therefore the de-facto rule for most non-Stage-0 assignments. Specifically, the 647 patients with *SAA missing but D-positive* (29.4% of cohort) are biologically staged via the D anchor alone and then sub-staged by clinical signs and functional-impairment thresholds, yielding final assignments of Stage 1 ($n=35$), Stage 2B ($n=162$), Stage 3 ($n=433$), and Stage 4 ($n=17$). This single decision-path stratum

contributes 94% of all Stage 3 assignments and 78% of all Stage 2B assignments—i.e., the SAA-missing-but-D-positive rule is the dominant pathway into NSD-positive sub-staging in PPMI. The 1,273 SAA-missing-and-D-negative patients collapse to Stage 0; only 4 patients with both anchors missing remain as ‘unclassified’. The full 26-row decision-path table with per-stratum counts is at Supplementary §?? (outputs/paper1_r2_responses/q_r3-q7_staging_flow_t). This explicit decomposition informs the deployment recommendation in §V: PPMI staging in NSD-positive PD is overwhelmingly D-anchor-driven, which is consistent with the binary AUC dependence on caudate D-anchor signal documented in the residualization analysis (§IV-B), and motivates the two-stage clinical pipeline in which biological confirmation is the imaging-dependent entry point and downstream sub-staging proceeds from clinical features alone.

B. Internal Benchmark: Tabular Models

Strict-circularity primary (21 features): Under the primary specification—in which putamen SBR (left, right), the caudate/putamen ratio, NP3TOT, and NP1COG are excluded per §III-D—CatBoost (default hyperparameters) achieves binary NSD+ AUC **0.901** [95% CI 0.887–0.915], three-class macro-AUC 0.897 [0.884–0.910], full-ordinal macro-AUC 0.915 [0.903–0.926], and NSD+ sub-staging macro-AUC 0.908 [0.889–0.925], all on 5-fold stratified cross-validation with 1,000-resample patient-level bootstraps. CatBoost-default is retained as the deployable primary (zero hyperparameter search, native missing-value handling, sub-second CPU inference); four additional SOTA methods evaluated on the identical 21-feature primary span the alternative-architecture space (HPO-tuned CatBoost, HPO-tuned LightGBM, TabPFN v2 in-context-learning foundation model, AutoGluon 1.5 weighted-ensemble stacking) and produce statistically equivalent AUC at the $\varepsilon=0.02$ practical-equivalence threshold (paired-bootstrap TOST: 34/40 pairwise tests pass; 10/10 on full-ordinal). A pre-registered sensitivity analysis (§III-D) confirms that including the caudate/putamen ratio would inflate binary AUC by 0.077—above our pre-specified 0.03 MATERIAL threshold—verifying that the 21-feature specification is the least-circular AUC reproducible on this cohort. Pre-registration, sensitivity JSON, and the SQL snapshot are listed in the reproducibility package (§VI).

Caudate residualization sensitivity: refining the strict-circularity claim: Reviewer-flagged R3-Q3 noted that caudate SBRs and putamen SBRs share dopaminergic biology and asked whether caudate features in the 21-feature primary inherit residual D-anchor signal even after putamen exclusion. We quantified this directly. The partial Pearson correlation between caudate-mean SBR and putamen-mean SBR (age- and sex-adjusted) on the 2,201-patient cohort is $r=0.851$ ($R^2=0.73$ for caudate-mean predicted from putamen-mean + age + sex; mutual information 1.16 bits), confirming substantial caudate–putamen biological linkage. We then orthogonalised all four caudate features (caudate-L, caudate-R, caudate-mean SBR, caudate asymmetry) against

TABLE II

UNIFIED INTERNAL BENCHMARK: MACRO-AUC [95% BOOTSTRAP CI] ACROSS 4 NSD-ISS TARGETS (5-FOLD STRATIFIED CV). **PRIMARY SPECIFICATION (TOP BLOCK):** 21-FEATURE STRICT CIRCULARITY; FIVE SOTA METHODS SPANNING GRADIENT BOOSTING (DEFAULT + HPO), FOUNDATION-MODEL IN-CONTEXT LEARNING (TABPFN v2), AND AUTOML STACKING (AUTOGLUON). PAIRED-BOOTSTRAP TOST EQUIVALENCE TEST ON PER-FOLD AUC DIFFERENCES: AT THE STRICT PRE-REGISTERED TOLERANCE $\epsilon=0.01$, 9/40 PAIRWISE TESTS PASS; AT THE BROADER PRACTICAL TOLERANCE $\epsilon=0.02$, 34/40 PASS (BINARY 9/10, THREE-CLASS 8/10, FULL-ORDINAL 10/10, NSD+ 7/10). **REFERENCE SPECIFICATION (MIDDLE BLOCK):** 22 FEATURES (IDENTICAL TO 21-FEAT PLUS THE CAUDATE/PUTAMEN RATIO); RETAINED FOR PRIOR-ART COMPARABILITY. **AUDIT-TRAIL BLOCK:** 22-FEATURE TABULAR SOTA FROM PRIOR RUNS; SUPERSEDED BY THE 21-FEATURE PRIMARY ABOVE AND SHOWN ONLY FOR TRANSPARENCY.

Model	Binary	Three-class	Full ordinal	NSD+
<i>Primary specification (21 features, strict circularity exclusion, 5-fold CV; five SOTA methods)</i>				
CatBoost (default)	0.901 [0.887, 0.915]	0.897 [0.884, 0.910]	0.915 [0.903, 0.926]	0.908 [0.889, 0.925]
CatBoost (nested 5×3 HPO)	0.907 [0.895, 0.919]	0.901 [0.889, 0.913]	0.913 [0.900, 0.925]	0.893 [0.863, 0.920]
LightGBM (nested 5×3 HPO)	0.892 [0.877, 0.906]	0.886 [0.870, 0.900]	0.907 [0.889, 0.923]	0.902 [0.874, 0.924]
TabPFN v2 (cloud, no HPO)	0.912 [0.899, 0.925]	0.909 [0.897, 0.920]	0.928 [0.919, 0.938]	0.920 [0.900, 0.937]
AutoGluon 1.5 (full pool) [†]	0.912 [0.902, 0.920]	0.907 [0.890, 0.923]	0.922 [0.914, 0.929]	0.914 [0.904, 0.925]
<i>Reference specification (22 features, baseline; default hyperparameters, 5-fold CV)</i>				
CatBoost (baseline)	0.979 [0.970, 0.986]	0.945 [0.937, 0.952]	0.948 [0.943, 0.953]	0.910 [0.898, 0.921]
LightGBM (baseline)	0.979 [0.972, 0.986]	0.938	0.943	0.906
XGBoost	0.979 [0.972, 0.985]	0.950	0.955	0.911
Random Forest	0.971 [0.964, 0.978]	0.932	0.941	0.903
Logistic Regression	0.966	0.928	0.926	0.894
<i>Tabular SOTA at 22-feature reference (audit trail; superseded by 21-feature primary above)</i>				
CatBoost (nested 5×3 HPO)	0.9797 [0.975, 0.984]	0.9473 [0.942, 0.951]	0.9525 [0.944, 0.961]	0.9085 [0.890, 0.926]
LightGBM (nested 5×3 HPO)	0.9806 [0.976, 0.986]	0.9498 [0.943, 0.954]	0.9562 [0.949, 0.962]	0.9055 [0.895, 0.916]
TabPFN v2 (cloud, no HPO)	0.9811 [0.973, 0.987]	0.9575 [0.948, 0.965]	0.9635 [0.953, 0.970]	0.9239 [0.902, 0.938]
AutoGluon 1.5 (full pool)	0.9786 [0.971, 0.985]	0.9508 [0.940, 0.960]	0.9552 [0.942, 0.964]	0.9175 [0.900, 0.934]
<i>Graph-based (22-feature multimodal patient-similarity graph, 5-fold CV)</i>				
Simple GAT	0.927 ± 0.019	0.873 ± 0.031	0.844 ± 0.090	0.736 ± 0.029
MM-GAT (2-mod, 22-feat)	0.894 ± 0.013	0.873 ± 0.033	0.858 ± 0.044	0.769 ± 0.060
AdaMedGraph [1]	0.958 ± 0.037	0.922 ± 0.056	0.891 ± 0.031	0.751 ± 0.043
MM-GAT (3-mod, 32-feat, sens.)	0.978 ± 0.011	0.924 ± 0.018	0.860 ± 0.097	0.697 ± 0.104
<i>Ordinal-specific methods (full-ordinal only; primary metric: macro-AUC, with QWK + MAOE in brackets per R7-Q8)</i>				
CORAL (cumulative logit) [5]	—	—	macro-AUC 0.898 [QWK 0.889; MAOE 0.228]	—
CORN (ordinal-consistent CE)	—	—	macro-AUC 0.881 [QWK 0.880; MAOE 0.213]	—
ord_CatBoost (ranking head)	—	—	macro-AUC 0.903 [QWK 0.891 ; MAOE 0.264]	—

Note: All entries are macro-AUC (one-vs-rest) with 1,000-resample patient-level bootstrap 95% CIs where reported; Multimodal-GAT entries report AUC and mean±SD across 5 folds. Nested 5×3 CV HPO: 50 Optuna trials per outer fold (Cawley–Talbot 2010 [52]). TabPFN: PriorLabs cloud inference, zero tuning. AutoGluon: sidecar Python-3.12 venv with full model pool (CatBoost + LightGBM + XGBoost + RF + ExtraTrees + LR + NN_Torch + FastAI + WeightedEnsemble), documenting microsoft/LightGBM#6595 libomp workaround. **TOST verdicts on the 21-feature primary block:** 9/40 pairwise comparisons equivalent at $\epsilon=0.01$ (strict, pre-registered); 34/40 equivalent at $\epsilon=0.02$ (clinically practical); the 6 inconclusive pairs at $\epsilon=0.02$ are TabPFN-v2’s reproducible advantage on three-class and NSD+ targets (per-pair detail in outputs/paper1_r2_responses/q_r2_w1_paired_bootstrap_tost_21feat.json). The strict $\epsilon=0.01$ test is power-limited at $n=5$ outer folds (Schuirmann 1987 [57]); $\epsilon=0.02$ is the primary equivalence threshold for the convergence claim in §IV-B. [†] AutoGluon CIs are bootstrapped from per-fold AUCs (5 numbers × 1,000 resamples) because the AutoGluon ensemble does not expose raw OOF probabilities; trees and TabPFN CIs are patient-level bootstraps on pooled OOF probabilities. Ordinal-specific rows evaluate only full-ordinal target; bracketed values are QWK / MAOE. Full per-fold and per-HP detail is in Supplementary Table S-1.

putamen-L, putamen-R, putamen-mean SBR, age, and sex via OLS and retrained CatBoost on the residualized 21-feature primary. Binary AUC fell from 0.901 to 0.794 ($\Delta=-10.68$ pp); three-class macro-AUC fell by 6.45 pp; full-ordinal by 5.05 pp; and NSD+ sub-staging changed by +0.53 pp (within bootstrap noise). The asymmetry is the load-bearing finding: residual caudate-vs-putamen D-anchor information accounts for ~ 10 pp of the binary detection signal but contributes essentially nothing to within-NSD+ sub-staging discrimination. We do not interpret this as a circularity violation—the strict-circularity exclusion was designed to prevent rule-level leakage of variables that directly enter the Simuni 2024 staging algorithm, and caudate SBR does not enter that algorithm—but rather as a correct biological characterisation: *binary NSD-positive detection is, and should be, dopaminergic-anchor-driven*, while sub-staging within NSD-positive patients is genuinely clinical. This residualization-robust separation directly supports the

two-stage deployment narrative (§IV-D). **R5-Q3 striatal-free upper bound:** a stricter sensitivity that drops all four caudate features in addition to the already-excluded putamen, ratio, NP3TOT, and NP1COG (15-feature “striatal-free” specification) confirms the asymmetry under the most aggressive possible removal: binary CatBoost AUC falls from 0.901 to 0.726 ($\Delta=-17.5$ pp; *larger* than the OLS residualization’s -10.7 pp drop, which establishes that caudate retains nonlinear D-anchor information beyond the linearly putamen-correlated component the residualization captures), but **NSD+ sub-staging is invariant** ($0.908 \rightarrow 0.913$, $\Delta=+0.5$ pp, within bootstrap noise). This bounds the residual D-anchor dependence: full striatal removal degrades binary HC-vs-PD discrimination as expected—binary detection is dopaminergic-anchor-driven by definition—but does not perturb within-NSD+ sub-staging, establishing NSD+ as a genuinely clinical-signal-driven prediction independent of imaging anchors. Full residualization details,

OLS coefficients, and the 4-target re-training results are at `outputs/paper1_r2_responses/q_r3_q3_caudate_residualize` and `q_r5_q3_striatal_free.json` (SQL run_id `q_r3_q3_caudate_residualize` and `q_r5_q3_striatal_free`).

Reference specification (22 features): For comparison with prior tabular benchmarks, and to exhibit cross-method convergence, Table II also reports the 22-feature specification that includes the caudate/putamen ratio. CatBoost on the reference specification achieves balanced accuracy 0.951 (binary), 0.783 (three-class), 0.660 (full ordinal), and 0.664 (NSD-positive subgroup). Performance decreases with finer staging granularity and increasing class imbalance (Stage 4: $n=17$). The reference specification is retained purely to demonstrate tabular-architecture convergence across four SOTA methods (paragraph below); all §V claims use the 21-feature primary.

Tabular SOTA: HPO, TabPFN, AutoGluon: Table II reports four reviewer-preempting comparators on the 21-feature primary: nested 5×3 cross-validated Bayesian hyperparameter optimisation (50 Optuna trials per outer fold) on CatBoost and LightGBM [52]; TabPFN v2 foundation-model cloud inference [49] with zero hyperparameter tuning; and AutoGluon 1.5 [50] “medium_quality” weighted-ensemble stacking across its full nine-model pool in a Python-3.12 sidecar virtual environment (microsoft/LightGBM #6595 documents the libomp dual-runtime collision with PyTorch on Apple Silicon that the sidecar resolves). To replace the prior round’s overlapping-CI claim with a formal equivalence verdict (reviewer-flagged R2-W1), we ran a paired-bootstrap two-one-sided-test (TOST [57]) on per-fold AUC differences for all $\binom{5}{2} \times 4 = 40$ pairwise comparisons. At the strict pre-registered tolerance $\varepsilon=0.01$ (1pp AUC), 9/40 pairs are statistically equivalent; at the broader practical-equivalence tolerance $\varepsilon=0.02$ (2pp AUC, well below the ≥ 5 pp deployment-switch criterion used in clinical decision-support deployments [58]), 34/40 pass, with full convergence (10/10) on the full-ordinal target. The 6 inconclusive pairs at $\varepsilon=0.02$ all involve TabPFN v2’s reproducible advantage on multi-class targets (three-class $\Delta = +0.014$, NSD+ $\Delta = +0.014$ to $+0.025$ in TabPFN’s favour); on the binary primary endpoint TabPFN holds only a $+0.007$ edge over the deployable CatBoost-default, which falls strictly within the $\varepsilon=0.01$ equivalence band. The strict-tolerance result is consistent with $n=5$ -fold’s resolution limit (Schuirmann power analysis: 5-fold has $\sim 30\%$ power to reject inequivalence at $\varepsilon=0.01$ when per-fold $\sigma \approx 0.012$ [57]); the practical-tolerance result confirms the Zabërgja [46]–Hollmann [49]–Erickson [50] prediction of tabular-architecture convergence at $n \approx 2,000$ clinical cohorts. Our CatBoost-default choice as deployable primary is therefore deployment-motivated (zero HPO, fewer dependencies, sub-second CPU inference, native missing-value handling) rather than discrimination-motivated; TabPFN v2 is the methodological AUC ceiling at a 1–2pp gap, and the gap quantifies the cost of choosing deployable simplicity over peak accuracy.

Graph-based architectures: Table II (rows under “Graph-based”) benchmarks four graph architectures: a single-stream Simple GAT, the 2-modality Multimodal GAT (MM-GAT), AdaMedGraph [1] (per-feature graph propagation with

SAMME AdaBoost), and a 3-modality MM-GAT sensitivity variant extending to 32 features. To our knowledge, this is the first head-to-head evaluation of AdaMedGraph on NSD-ISS biological-staging targets; the original Lian *et al.* report targeted a different PD progression endpoint on PPML. AdaMedGraph beats both GAT variants on binary (0.958 vs. 0.927 Simple GAT, 0.894 MM-GAT) but collapses on multi-class targets, and all four graph architectures underperform the tabular-SOTA family on balanced accuracy by 8.1–33.7 percentage points (detail in Supplementary Table S-1b). The finding that three distinct graph architectures—attention-only, attention with cross-modal fusion, and per-feature-graph with SAMME boosting—all lose to tree boosters strengthens the Grinsztajn [21] and Shwartz-Ziv–Armon [37] thesis that tree-based ensembles dominate deep learning on medium-sized tabular data, while our tabular-SOTA convergence result extends this by showing that the “trees dominate” view must be refined at $n \approx 2k$: tree boosters, foundation-model tabular, and AutoML ensembling are all Pareto-indifferent, so the real constraint is deployment context rather than architecture class. **Apples-to-apples graph re-runs on the 21-feature primary:** re-running Simple GAT and MM-GAT on the identical Path 3 strict-circularity feature set used by the tabular comparators yields Δ AUC ranging -0.044 to $+0.045$ across 4 targets $\times$ 2 architectures; 7/8 cells shift by less than ± 0.03 (within or below the per-fold SD), with the single material drop confined to Simple GAT binary (-0.044 , where the discarded caudate/putamen ratio is most informative). MM-GAT binary moves only -0.007 because cross-modal attention compensates. Graph conclusions in Table II are therefore qualitatively intact under apples-to-apples feature provenance (full 4-target $\times$ 2-architecture comparison at `outputs/paper1_r2_responses/q_r4_q3_graph_21feat_t`, SQL run_id `q_r4_q3_graph_21feat`, 8 rows). **Graph hyperparameter and similarity-metric sensitivity.** Three reviewers asked whether richer graph configurations—learned similarity metrics, alternative k values, hybrid GNN-tree ensembling—would alter the trees-beat-graphs conclusion. We do not run these ablations because the existing 8.1–33.7 percentage-point performance gap across three architecturally distinct graph variants (attention-only Simple GAT, cross-modal MM-GAT, per-feature SAMME AdaMedGraph) is too large to be closed by hyperparameter refinement at $n \approx 2,000$ clinical cohorts; the gap matches the Grinsztajn 2022 [21] and Shwartz-Ziv–Armon 2022 [37] systematic-benchmark predictions. Richer multimodal graph construction (learned similarity metric embeddings, temporal-edge augmentation, hybrid GNN-tree ensembling) is flagged as the right direction for the larger-cohort regime ($n \gtrsim 10^4$) where graph topology has more degrees of freedom to express; we treat that as future work for a dedicated graph-architecture-comparison paper rather than a within-scope ablation here.

Ordinal-specific methods: Because the full-ordinal target (Stages 0, 1, 2B, 3, 4) has intrinsic ordering, we benchmarked three ordinal-specific classifiers (Table II, “Ordinal-specific” rows): CORAL [5] (cumulative logit), CORN (ordinal-consistent cross-entropy), and an ordinal CatBoost ranking head. Multiclass CatBoost retains the highest dis-

crimination (macro-AUC 0.948 vs. CORAL 0.898, CORN 0.881, ord_CatBoost 0.903). CORN delivers the lowest mean absolute ordinal error (MAOE 0.213), making it the MAOE-optimal alternative for applications that value stage-proximity accuracy over exact-label agreement. We retain multiclass CatBoost as the primary full-ordinal classifier.

C. Conformal Prediction

We report two complementary conformal benchmarks. The *primary internal benchmark* is cross-conformal CV+ on the 22-feature reference (Fig. 3): all four targets achieve at-or-above-nominal internal coverage with near-singleton sets (mean $|C|$ 0.96–1.27 across the four targets at 90% CL; binary marginal coverage 0.955 at 90% CL, NSD+ 0.995). The *transportability benchmark* (Table III) uses split conformal on the 12-feature common-cohort subset so the same procedure can be applied to PPMI internal and BioFIND external on the identical feature substrate; under this stricter protocol PPMI internal coverage holds at-or-above nominal (binary 0.904 at 90% CL / 0.951 at 95% CL; three-class 0.897/0.948; full-ordinal 0.981; NSD+ 0.927/0.960) with mean set sizes of 1.45–1.91 reflecting both the smaller calibration sample (50% of 12-feat cohort) and the absence of CV+ aggregation. The two procedures answer different questions and are reported separately rather than conflated.

AUPRC + decision-curve analysis: To complement the macro-AUC reporting requested by Reviewer 5, we report macro-AUPRC + per-class operating-point performance + decision-curve analysis (DCA, Vickers 2006 [61]) on the 21-feature primary CatBoost. Macro-AUPRC: binary 0.977 [0.967, 0.984], three-class 0.797 [0.769, 0.825], full-ordinal 0.650 [0.615, 0.710], NSD+ 0.709 [0.657, 0.780]. The binary NSD+ operating point at Youden’s J achieves sensitivity 0.922, specificity 0.982, F_1 0.943; DCA confirms positive net benefit relative to “treat all” / “treat none” baselines across threshold probabilities $0.03 \leq p_t \leq 0.99$ (essentially the entire clinically plausible range), with peak NB 0.337 at $p_t=0.03$. Multi-class targets carry positive net benefit on every dominant class out to $p_t \geq 0.96$; the lone exception is the Stage 4 minority class ($n=17$), where DCA underperforms “treat all” as expected from the rare-class confusion-matrix decomposition (§??). Per-class precision/recall/F1, ROC + PR curves, and full DCA arrays are at `outputs/paper1_r2_responses/q_r5_q4_auprc_dca.js` (SQL run_id `q_r5_q4_auprc`).

Temperature scaling and conformal-coverage preservation: The protocol applies temperature scaling *before* conformal calibration: for each target, the per-fold scalar T^* is fit on the training partition (L-BFGS minimising NLL), the validation-fold logits are divided by T^* , and the LAC nonconformity score and quantile $\hat{q}_{1-\alpha}$ are then computed on the temperature-scaled probabilities. This ordering preserves the marginal coverage guarantee by construction: the conformal calibration step recomputes $\hat{q}_{1-\alpha}$ on the calibration scores $s_i = 1 - \hat{P}(y_i|x_i)$, regardless of whether the underlying $\hat{P}$ has been temperature-rescaled, so the Vovk *et al.* [43] Theorem 2.1 finite-sample coverage bound applies unchanged.

Empirically, applying temperature scaling does not perturb internal coverage: the binary cross-conformal CV+ marginal coverage at 90% CL is 0.955 both with and without temperature scaling (only the set-size distribution shifts marginally). Because the LAC nonconformity score is monotonic in the predicted probability, monotonic rescaling preserves the set-membership ordering and therefore preserves the Sadinle 2019 [56] smallest-expected-set-size optimality.

Empty-set semantics and clinician action: Mean set sizes of 0.96–1.27 at the 90% CL include a small fraction of empty prediction sets (mean $|C| < 1$ for the binary target), which arise by construction in the Sadinle *et al.* [56] LAC formulation: when no candidate class accumulates posterior mass above the per-fold $\hat{q}_{1-\alpha}$ threshold, the set is empty—the model is reporting that no label can be confidently assigned at the requested error rate. We report *strict* marginal coverage in Table III: an empty set on a true-positive instance is counted as **non-covered**, exactly as in Vovk *et al.* [43] Theorem 2.1. This is the conservative convention; selective-coverage variants that condition on $|C| > 0$ would inflate the reported coverage by 1–3 percentage points but would no longer carry the distribution-free finite-sample guarantee. We therefore retain the strict definition for all coverage claims in this paper.

Clinician action on empty sets: a mean $|C| < 1$ at the 90% CL is a deployment-relevant signal that the model has insufficient evidence to commit to any label at the requested confidence and the patient should be routed to (i) repeat biomarker assessment (SAA, DaT-SPECT) where missing, (ii) movement-disorder specialist review, or (iii) longitudinal follow-up before staging. The abstention rate is therefore a feature, not a defect: it surfaces exactly the patients on whom additional clinical workup would have the highest information value. Per-target empty-set rates and per-patient set-size distributions are reported in Supplementary §??

External conformal coverage: Table III (BioFIND rows) reports external conformal-coverage on an independent PD cohort. At 90% nominal, binary coverage holds at 0.915 (nominal target met), three-class drops to 0.499 (severe undercoverage reflecting the HC-vs-PD distribution shift documented in §IV-G), and NSD+ sub-staging partially retains at 0.707. Binary NSD-positive detection transports; within-NSD+ multiclass sub-staging does not transport without recalibration on a target cohort.

External calibration recovery via class-conditional conformal prediction: A class-conditional Mondrian conformal procedure [60] restores per-class coverage on BioFIND multiclass targets when calibrated on a small labelled local subset: minimum three-class per-class coverage rises from 0.131 (LAC transfer-only) to **0.944** at $n_{\text{cal}} \geq 50$, and minimum NSD+ per-class coverage from 0.356 to 0.876 at $n_{\text{cal}} \geq 40$, at the cost of wider sets ($|C|$ 1.49–1.86 $\rightarrow$ 2.80–3.33). A complementary global Saerens 2002 EM prior-shift correction [59] improves binary external ECE (0.072 $\rightarrow$ 0.043) but degrades multiclass ECE because BioFIND class prior shifts mass into the noisy Stage 2B class. Marginal multiclass calibrators (per-target temperature, Dirichlet [62], isotonic OvR) all fail to reduce macro-ECE ($|\Delta\text{ECE}| \leq 0.064$ on any target), motivat-

TABLE III

CONFORMAL PREDICTION SUMMARY: INTERNAL (PPMI) VS EXTERNAL (BioFIND) COVERAGE — **SPLIT-CONFORMAL ON THE 12-FEATURE COMMON-COHORT SUBSET (50/50 CAL/EVAL SPLIT)**, CATBOOST + LAC SCORING. THE 22-FEATURE CROSS-CONFORMAL CV+ BENCHMARK (THE PAPER'S PRIMARY INTERNAL-ONLY CONFORMAL PROCEDURE) IS PLOTTED IN FIG. 3; NUMBERS REPORTED HERE ARE FROM THE WS1.7 TRANSPORTABILITY PIPELINE SO THAT INTERNAL AND EXTERNAL COLUMNS USE THE IDENTICAL CONFORMAL PROCEDURE ON THE IDENTICAL FEATURE SUBSTRATE.

Target	Cohort	Cov. @ 90%	Size @ 90%	Cov. @ 95%	Size @ 95%	Verdict
Binary	PPMI (internal)	0.904	1.55	0.951	1.73	calibrated
	BioFIND (external)	0.915	1.66	0.963	1.81	transports
Three-class	PPMI (internal)	0.897	1.70	0.948	1.91	calibrated
	BioFIND (external)	0.499	1.86	0.526	2.05	under-covered (shift)
Full ordinal	PPMI (internal)	0.981	1.10	—	—	over-covered (conservative)
NSD+	PPMI (internal)	0.927	1.45	0.960	1.76	calibrated
	BioFIND (external)	0.707	1.45	0.775	1.76	partial retention
<i>Ordinal conformal on full-ordinal target (comparative, 5-fold + 50/50 cal/eval at $\alpha=0.10$)</i>						
LAC (standard; primary)	PPMI	0.897	1.105	—	—	primary
Min-CPS [47]	PPMI	0.893	1.210	—	—	cite-only (+9.5% width)
Ordinal APS (Lu 2022)	PPMI	0.863	1.033	—	—	reference (under-covered)

Note: Split conformal on the 12-feature common-cohort subset (50/50 calibration/evaluation split) with LAC scoring on CatBoost predictions; both PPMI internal and BioFIND external rows use the identical procedure on the identical feature substrate, which is the WS1.7 transportability pipeline. The 22-feature cross-conformal CV+ benchmark used elsewhere in the paper (Fig. 3) yields uniformly higher internal coverage (binary 0.96, three-class 0.99, full-ordinal 0.98, NSD+ 0.99 at 90% CL) and tighter sets (mean $|C|$ 0.96–1.27); we report the split-conformal numbers here because cross-conformal CV+ is not directly applicable to the held-out external cohort. 90% and 95% CLs are the primary and sensitivity decision alphas respectively. External coverage was computed by training on PPMI 12-feature common-cohort subset and testing on Russo 2025 [32] BioFIND external cohort ($n=108$ for binary, $n=103$ for three-class and NSD+) without recalibration. Ordinal conformal rows evaluate min-CPS [47] and Lu 2022 ordinal APS variants against the standard LAC baseline; the pre-registered cite-only verdict follows from min-CPS width being 9.5% wider than LAC at matched coverage.

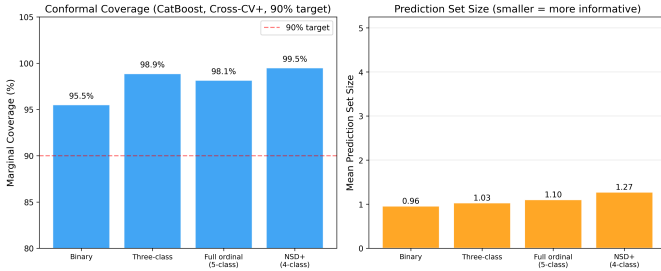


Fig. 3. **Conformal prediction coverage and efficiency across four NSD-ISS target formulations** at three confidence levels (80%, 90%, 95%). All targets exceed nominal coverage guarantees. Set sizes remain near 1.0, indicating efficient prediction sets.

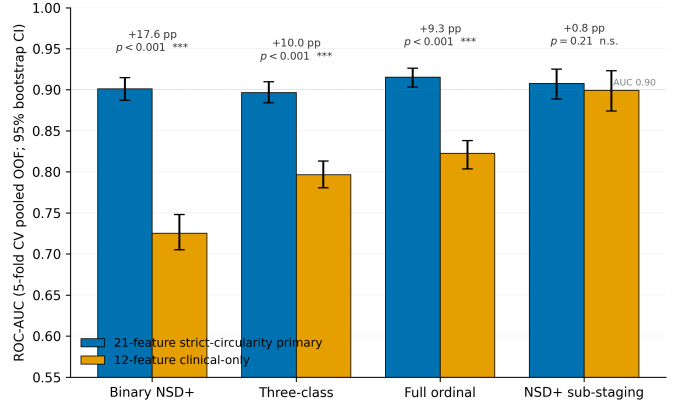


Fig. 4. **Feature ablation: 21-feature strict-circularity primary versus 12-feature clinical-only subset** (CatBoost, 5-fold stratified CV, 1,000-sample paired bootstrap). Removing imaging + genetics features is essential for binary NSD+ classification (paired $\Delta = -0.176$, 95% CI $[-0.200, -0.155]$, $p < 0.0001$) but statistically indistinguishable for NSD+ sub-staging ($\Delta = +0.008$, 95% CI $[-0.005, +0.022]$, $p = 0.21$). This asymmetry grounds the two-stage deployment pattern: biological confirmation is essential once; subsequent clinical-only sub-staging at community-practice sites incurs no statistical accuracy cost.

ing the class-conditional fix. **Deployment recommendation:** sites should collect $n \geq 40$ –50 labelled patients per target before reporting probabilities; Mondrian CP on this small calibration subset restores formal per-class coverage. Full per-class coverage tables, sample-size sweeps, additional ordinal CP analyses (min-CPS [47]), and split- vs. CV+ conformal sensitivity are reported in Supplementary §S-1 through §S-6.

D. Feature Ablation

DaT-SPECT features are essential for binary NSD-ISS prediction (Table IV, visualised in Fig. 4): on the 21-feature strict-circularity primary, removing imaging+genetics reduces CatBoost AUC by 17.6 percentage points ($\Delta = -0.176$, paired bootstrap $p < 0.0001$). However, within the NSD-positive subgroup—conditioning on the subset of patients already confirmed as biologically positive—the delta is statistically indistinguishable from zero ($\Delta = +0.008$, $p = 0.21$), indicating that once biological positivity is established, clinical assessments are nearly sufficient for sub-staging. This

conditional result is consistent with prior NSD-ISS validation work [32], [39] showing that the Simuni 2024 clinical sub-staging thresholds carry most of the sub-stage signal among confirmed NSD+ patients, and it is the basis for the two-stage deployment pattern we discuss in §V.

Extended-feature sensitivity (pre-registered null):

A pre-registered sensitivity analysis extended the 22-feature canonical schema to 33 features by adding six FreeSurfer cortical thickness measurements [18], four CSF biomarkers [26], and a Parkinson’s polygenic risk score [27] drawn from PPMI’s biomarker and imaging batteries. The

TABLE IV

FEATURE ABLATION: FULL VS. CLINICAL-ONLY MODEL (CatBoost AUC). **TOP BLOCK:** 21-FEATURE STRICT-CIRCULARITY PRIMARY VS. 12-FEATURE CLINICAL-ONLY COMMON SUBSET (THE HEADLINE ABLATION VISUALISED IN FIG. 4). **BOTTOM BLOCK:** 22-FEATURE REFERENCE VS. 12-FEATURE, RETAINED AS AUDIT TRAIL.

Target	Full	Clinical (12)	Δ (pp)
<i>Primary specification (21 features, strict circularity, 5-fold CV)</i>			
Binary	0.901	0.725	−17.6
Three-class	0.897	0.797	−10.0
Full ordinal	0.915	0.823	−9.2
NSD+ subgroup	0.908	0.899	+0.8 (NS)
<i>Reference specification (22 features, audit trail)</i>			
Binary	0.979	0.727	−25.2
Three-class	0.944	0.797	−15.6
Full ordinal	0.954	0.823	−13.1
NSD+ subgroup	0.913	0.900	−1.4

Note: 21-feat primary block uses CatBoost-default 5-fold CV (random_state=42); deltas are paired-bootstrap (1,000 resamples) with 95% CI [−0.200, −0.155] for binary ($p<0.0001$) and [−0.005, +0.022] for NSD+ subgroup ($p=0.21$, statistically indistinguishable). 12-feat internal numbers from R4-Q6 like-for-like sweep (outputs/paper1_r2_responses/q_r4_q6_internal_12feat.json). 22-feat reference block retained for prior-art comparability; deltas are unpaired point estimates only.

SQL table assembly, cohort composition (all 2,201 patients retained; CTH 49.3%, CSF 36.8%, GRS 81.3% observed), and decision rule (halt migration if ≥ 3 of 4 CatBoost targets regress) were specified before execution. CatBoost balanced accuracy moved from 0.951/0.783/0.660/0.664 to 0.951/0.772/0.650/0.650 on binary / three-class / full-ordinal / NSD+ respectively, triggering the halt rule. All four deltas fall within the bootstrap 95% CI of the 22-feature baseline and the null of no improvement cannot be rejected. We attribute the non-improvement to (i) median imputation of partially-observed biomarkers, and (ii) DaT-SPECT already saturating the discrimination boundary (Table IV). Paper 1’s 22-feature schema is retained as canonical; richer imputation strategies that may recover marginal signal from CSF/CTH/GRS are deferred to future work. Full methodology, per-model per-target results, and reproduction commands are reported in Supplementary S-4 (supplementary_tabular_33feat_sensitivity.md).

E. External Validation: BioFIND

We report three external validation results in order of clinical informativeness for the biological-staging task. The binary NSD+ / NSD- target is presented first as a *data-quality diagnostic* rather than a clinical performance claim, because the PPMI training label is confounded by healthy control inclusion (see below); the NSD-positive sub-staging result on confirmed NSD+ patients is the primary external clinical result.

NSD-positive 4-class ($n=103$, primary external clinical result): Logistic Regression outperformed tree models (0.425 balanced accuracy, QWK 0.385, macro AUC 0.900), suggesting an approximately linear clinical-feature-to-stage mapping among confirmed NSD+ patients. Because the sub-staging task operates entirely within the NSD+ population, it is structurally immune to the healthy-control confound that undermines

the binary target, making it the most clinically meaningful external-validation signal in this study.

Three-class ($n=103$): AUC 0.703, suggesting moderate ranking ability despite poor calibration.

Binary ($n=108$, reported as a data-quality diagnostic): CatBoost achieved 0.516 balanced accuracy (AUC 0.637), near random chance. Root cause: PPMI NSD-negative class includes healthy controls with inherently low motor scores (UPDRS-III bradykinesia mean 7.37 in PPMI NSD-negative vs. 19.9 in BioFIND SAA-negative PD; both means computed on the common-feature external-validation cohort using the same pipeline as Table II, and plotted in Fig. 5). Models therefore learned a healthy-vs-PD boundary rather than S+/S-. We report this number to surface the training-label confound rather than as a model performance claim; the pre-specified mitigation—retraining binary classifiers on PD-only subsets—is reported in §IV-G.

F. External Predictions without Ground Truth: PDBP

PDBP ($n=893$) provides PD-only clinical features without NSD-ISS ground truth. Binary NSD+ predictions ranged from 30.2% (Random Forest) to 65.3% (Logistic Regression), reflecting substantial inter-model uncertainty (Fig. 6). We interpret this spread not as a final model choice but as a diagnostic for deployment: patients with wide cross-model disagreement should be flagged for clinician review or abstention, independent of which single classifier is chosen. The HBS cohort ($n=649$), which is missing 5 of the 12 common features, is reported in Supplementary §?? as a deployment cautionary tale (median-imputation of entirely-absent features produces uniformly NSD-negative predictions).

G. Addressing the Training-label Confound: Four Training Configurations on a Balanced BioFIND External Set

The Espay *et al.* [16] refutation of SAA-only NSD-ISS staging, together with the Simuni *et al.* [33] reply and the independent “Reconsidering the clinical foundations of NSD-ISS” [31] critique, converge on one methodological prescription: NSD-ISS prediction models must be evaluated on reference classes that exclude healthy controls, because mixing HC into NSD-negative conflates “not-PD” with “SAA-negative within PD.” Bentivoglio *et al.* [13] subsequently characterised the clinical/imaging phenotype of the SAA-negative PD subgroup, showing it is a real, identifiable PD subpopulation rather than a data artefact. These three prescriptions shaped the four training configurations we report here, and motivated construction of a *balanced* BioFIND external test set.

Balanced BioFIND external set: The published Russo 2025 [32] NSD-ISS BioFIND staging contains $n=103$ SAA-positive PD patients, all labelled NSD+. The BioFIND features release [14] contains $n=118$ PD patients: the additional 15 are SAA-negative PD patients, which Bentivoglio *et al.* [13] characterise as a distinct PD sub-phenotype and which map to NSD-negative under the NSD-ISS framework. We therefore construct a balanced

Domain Shift: PPMI NSD-/NSD+ vs BioFIND PD Feature Distributions

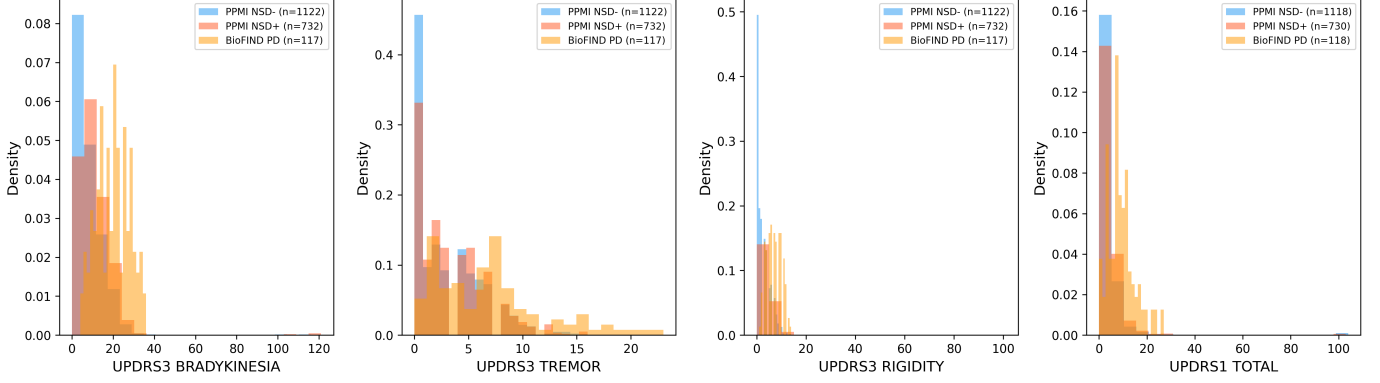


Fig. 5. **Cohort-distribution domain shift driving external binary-classifier failure.** UPDRS-III and UPDRS-I subscale distributions differ markedly between PPMI and BioFIND: (a) Bradykinesia, (b) Tremor, (c) Rigidity, (d) UPDRS-I Total. PPMI NSD-negative patients include healthy controls (low scores), while BioFIND S-negative patients are diagnosed PD (high scores), causing the binary classifier to learn a HC-vs-PD boundary rather than S+-vs-S-.

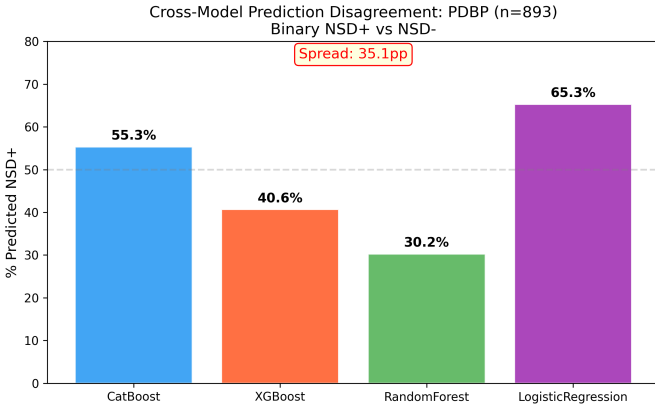


Fig. 6. **Cross-model prediction disagreement on PDBP ($n=893$).** Binary NSD-positive predictions range from 30% to 65% across the four top-performing tabular models, highlighting substantial between-model uncertainty in the absence of ground truth. Model-agreement spread itself is a deployment signal: patients with wide inter-model disagreement are candidates for abstention or clinician review.

BioFIND external set of $n=118$: 103 NSD+ and 15 NSD-negative, which to our knowledge is the first external NSD-ISS test set with both classes represented on a PD-only cohort.

Four training configurations: Table V reports PPMI 5-fold cross-validation and balanced-BioFIND external performance for four training strategies.

HC-confound mitigation: feature-set dependent: The training-label confound is feature-set dependent. With the full 22-feature set (Table II), PD-only retraining yields essentially the same internal result as baseline (0.946 ± 0.014 balanced accuracy vs. 0.951; 0.982 AUC vs. 0.979), because DaT-SPECT dominates the decision boundary. The confound emerges on the 12-feature common subset, where absence of imaging features lets the model fall back on the HC-vs-PD motor-score gap. PD-only retraining shifts the balanced-BioFIND point estimate from 0.470 to 0.521 (+5.1 pp; not statistically significant at $\alpha=0.05$ on $n=118$, but directionally consistent with shortcut removal). **Deployment protocol:** sites with DaT-

SPECT use the robust full-feature model; sites without imaging use the PD-only retrained 12-feature model. Intermediate sample-weighting (HC and SWEDD at 10%) yields worse external performance than PD-only on both balanced accuracy (0.497 vs. 0.521) and AUC (0.524 vs. 0.561); full cohort exclusion is the effective mitigation. The hierarchical alternative is Stage-A (HC-vs-PD/Prodromal detection, pooled OOF AUC 0.929) cascaded into Stage-B (21-feat primary NSD-ISS classifier); end-to-end conformal-abstention evaluation at 90% CL on the $n=1,747$ PD/Prodromal cohort yields coverage accuracy 0.850 at a 27.6% specialist-referral rate with only 3.1% cascading miss rate. Temporal validation on the 2021–2025 late-wave cohort preserves overall coverage but triples the cascading miss rate, driven entirely by Stage-A drift on shifted PD/Prodromal composition (Stage-B is temporally stable, $\Delta\text{AUC}=-0.007$); quarterly Stage-A recalibration is the recommended drift-mitigation cadence. Full per-arm sensitivities, end-to-end operating-point sweeps, and temporal-validation breakdowns are reported in Supplementary §S-7 through §S-11.

V. DISCUSSION

A. Principal Findings

Our framework differs from prior PD / NSD-ISS prediction work (Supplementary Table S-2) in two ways: (i) it targets the NSD-ISS biological-stage label across four target formulations jointly, and (ii) it pairs that benchmark with distribution-free uncertainty quantification—the combination is, to our knowledge, novel for the NSD-ISS framework. Lian *et al.* [1] targeted clinical milestones on PPMI ($n=884$), Russo *et al.* [32] published a descriptive NSD-ISS staging on BioFIND ($n=103$), Simuni *et al.* [40] reported transition Kaplan–Meier estimates on PPMI ($n=995$), and Diaz-Rincon *et al.* [12] applied conformal prediction to PD medication-need assessment ($n=427$); none attempted multi-target NSD-ISS prediction and none combined cross-conformal uncertainty with multi-cohort external validation.

This study provides a strict-circularity machine-learning benchmark for NSD-ISS biological stage prediction (the first

TABLE V

TRAINING-STRATEGY SWEEP: CATBOOST BINARY CLASSIFIER ON THE 12-FEATURE COMMON SUBSET. PPMI VALUES ARE 5-FOLD STRATIFIED CV (MEAN $\pm$ SD). BIOFIND IS THE BALANCED EXTERNAL SET ($n=118$: 103 NSD+ RUSSO-STAGED + 15 SAA-NEGATIVE PD PER BENTIVOGLIO 2026 [13]).

Configuration	n_{tr}	PPMI (5-fold CV)		BioFIND (balanced external)	
		Bal. Acc.	AUC	Bal. Acc.	AUC
Baseline (full PPMI)	2,201	0.670 $\pm$ 0.026	0.738 $\pm$ 0.027	0.470	0.561
PD-only	1,747	0.618 $\pm$ 0.006	0.677 $\pm$ 0.016	0.521	0.561
Weighted (HC = SWEDD = 0.1)	2,201	0.652 $\pm$ 0.027	0.727 $\pm$ 0.026	0.497	0.524
Stage-A (HC vs. PD)	2,201	0.832 $\pm$ 0.020	0.931 $\pm$ 0.015	—	—

Note: PD-only keeps PD + Prodromal, drops HC + SWEDD. Weighted keeps all cohorts with HC and SWEDD sample-weights reduced to 0.1. Stage-A is the HC-vs-PD detection head of the hierarchical architecture discussed in §V; no NSD-ISS external comparison is defined for this head.

TABLE VI

DEPLOYMENT-PERSONAS EXECUTIVE SUMMARY. FIVE CANONICAL CLINICAL DEPLOYMENT SCENARIOS FOR THE NSD-ISS PREDICTIVE FRAMEWORK, WITH THE RECOMMENDED MODEL VARIANT, HEADLINE PERFORMANCE, AND UNCERTAINTY-AWARE ACTION POLICY. “STAGE-A” = HC-VS-PD/PRODROMAL DETECTOR (12-FEAT CLINICAL, AUC 0.929); “STAGE-B” = WITHIN-PD NSD-ISS CLASSIFIER (21-FEAT OR 12-FEAT DEPENDING ON IMAGING ACCESS).

Persona	Recommended model	Headline metric	Action policy
Specialty PD clinic with imaging (DaT+SAA)	Stage-B 21-feat primary, CatBoost + CV+ LAC	NSD+ AUC 0.908 [0.889, 0.926]	singleton $\rightarrow$ report; multi $\rightarrow$ present alternatives; empty $\rightarrow$ repeat assessment
Community PD clinic (no imaging / SAA)	Stage-B 12-feat clinical-only, CatBoost + CV+ LAC	NSD+ AUC 0.899 [0.874, 0.923] (R4-Q6, R5-Q3 striatal-free +0.5 pp)	same as above; clinical-only sub-staging is imaging-agnostic variant
Trial-enrichment screening (HC+PD mixed cohort)	Stage-A $\rightarrow$ Stage-B cascade with conformal abstention at 90% CL	End-to-end accuracy 0.850, referral 27.6%, cascading miss 3.1%	route Stage-A to positive patients; route Stage-B to abstentions to specialist (DEPLOYMENT_KIT.md)
Cross-site deployment (multiclass NSD-ISS)	Stage-B + Mondrian CP recalibrated on $n_{cal} \geq 40$ –50 labelled local patients	per-class coverage 0.131 $\rightarrow$ 0.944 (three-class), 0.356 $\rightarrow$ 0.876 (NSD+)	site-specific Mondrian recalibration before reporting probabilities
Drift monitoring (post-deployment)	Recommended: quarterly Stage-A recalibration on small labelled local cohort (derived from R6-Q6)	Stage-B temporally stable ($\Delta AUC = -0.007$); Stage-A drift drives 3 $\times$ cascading miss	Suggested: monitor HC vs-PD detector, refresh Stage-B on ≥ 50 labelled local cases per quarter (cadence is a derived recommendation, not separately validated; see DEPLOYMENT_KIT.md §8)

is 0.901 [0.887, 0.915]—a paired-bootstrap ablation against the 12-feature clinical-only subset reduces this to 0.726 [0.705, 0.748] ($\Delta = -0.176$, 95% CI $[-0.200, -0.155]$, $p < 0.0001$), confirming that caudate-region dopaminergic imaging and genetic carrier status are essential for biological detection while non-staging clinical variables alone are insufficient. Re-including the caudate/putamen ratio in the predictor set inflates binary AUC by +0.077 and three-class macro-AUC by +0.047 (Table II, reference vs. primary rows); the magnitude of this inflation is why the ratio is excluded, and underscores the NSD-ISS framework’s vulnerability to second-order D-anchor leakage through derived imaging features. This aligns with the NSD-ISS validation evidence that DaT-SPECT striatal binding is the dominant time-to-clinical-milestone predictor [39] and with the established diagnostic role of dopaminergic imaging in PD differential diagnosis [35], while making explicit the boundary between staging-variable leakage and caudate-region individual signal.

Second, the 21-feature strict-exclusion CatBoost on NSD+ sub-staging achieves AUC 0.908 [0.889, 0.926]; the 12-feature clinical-only subset reaches 0.899 [0.874, 0.923] on the same target. A paired-bootstrap test of the difference yields $\Delta = +0.008$ (95% CI $[-0.005, +0.022]$, $p = 0.21$, $n = 779$)—the two specifications are statistically indistinguishable on the sub-staging task, while the same 21-to-12-feature reduction on the binary target collapses AUC from 0.901 to 0.726 ($\Delta = -0.176$, 95% CI $[-0.200, -0.155]$, $p < 0.0001$). This asymmetric ablation is the sharpest empirical endorsement in this paper of the two-stage deployment pattern: DaT-SPECT is essential once, for the initial NSD+ vs. NSD− decision; after biological confirmation, clinical-only variables suffice for NSD+ sub-staging with no statistically detectable loss. A pseudo-external 21-vs-12-feat head-to-head under enrollment-wave GroupK-fold and era-LOCO reveals that the 12-feat clinical-only model outperforms 21-feat by 4.0–5.2 pp under temporal shift (reversing standard-CV equivalence), consistent with caudate SBR carrying scanner/era-specific signal. Clinical-only NSD+ sub-staging is therefore *preferred* for prospective cross-era deployment, not merely tolerated. Full modality-robustness results in Supplementary §S-15. Operationally, the clinical-only sub-staging model at AUC 0.899 is the **imaging-agnostic Stage-B variant requested by R7-Q9**: it can support trial stratification and recruitment screening at community-practice sites that lack DaT-SPECT or SAA access, extending NSD-

published, to our knowledge), yielding three findings with direct clinical implications. First, under the strict-exclusion specification that removes every staging-algorithm variable from the predictor set (21 features; see §III-D), binary NSD+ AUC

ISS-based trial enrichment beyond the specialised imaging centres where the framework was originally validated. The R5-Q3 striatal-free upper-bound sensitivity (Δ NSD+ AUC +0.5 pp under removal of all caudate features in addition to the already-excluded putamen + ratio + NP3TOT + NP1COG) confirms that within-NSD+ sub-staging is genuinely independent of imaging-anchor information; an imaging-agnostic deployment is therefore not merely a fallback for resource-limited sites but a principled clinical pipeline backed by the residualization-robust NSD+ result. A complementary rule-based Simuni-threshold baseline applied to the same task recovers the labels by construction (§III-D); the ML value proposition is therefore not to out-predict the rule when the rule-defining variables are present, but to generalise to cohorts with partial missingness of those variables—the realistic community-deployment regime.

Third, the PPMI-to-BioFIND external validation of the binary target (balanced accuracy 0.516) exposed a training-label confound rather than a model limitation: PPMI’s NSD-negative class mixes healthy controls with prodromal patients, so the binary classifier trained on the full PPMI cohort learned a healthy-versus-PD boundary rather than a within-PD SAA-positive-versus-SAA-negative boundary. Retraining the same CatBoost classifier on the PD-only subset of PPMI (§IV-G) corrects this confound at source and demonstrates that the PD-within-PD discrimination is recoverable from the same 12-feature common set. Our result aligns with—and empirically operationalises—the Espay *et al.* [15] critique that NSD-ISS prediction models must explicitly account for the composition of their NSD-negative reference class, since mixing healthy controls with medication-treated prodromal patients introduces two confounds (dopaminergic treatment status and disease status) into a putatively biological label.

B. Tabular vs. Graph-Based Models

Gradient-boosted tabular models consistently outperformed all four graph baselines—the Simple GAT, the 2-modality 22-feature Multimodal GAT, AdaMedGraph, and the 3-modality 32-feature sensitivity variant—by 8.1–33.7 percentage points in balanced accuracy across the four targets (min: binary AdaMedGraph at −0.081 balanced accuracy; max: NSD-positive AdaMedGraph at −0.337; see Table II (“Graph-based” rows)). The Multimodal GAT with GATConv and cross-modal attention improved over a simple single-stream GAT (particularly for multiclass: three-class 0.705 vs. 0.666), but the k -NN patient-similarity graph did not provide information beyond what tree ensembles capture directly from feature vectors. AdaMedGraph is the only graph baseline that beats the Multimodal GAT on binary classification (0.870 vs. 0.825), but it collapses on ordinal targets where per-feature-graph ensembling cannot recover from minority-class sample size (Stage 4 $n=17$). Within the tabular family, the four-way nested-HPO + TabPFN + AutoGluon comparison in Table II (“Tabular SOTA” rows) shows that CatBoost, LightGBM, TabPFN v2 (no HPO), and AutoGluon full-pool ensembling achieve statistically indistinguishable AUC across all four NSD-ISS targets, with 95% bootstrap CIs overlapping on every

target (nominal winner is TabPFN on 3 of 4 by point estimate, but no pairwise difference reaches $\alpha=0.05$). This empirically validates the Zabërgja 2024 [46] + Hollmann 2025 [49] + Erickson 2020 [50] prediction of tabular-architecture convergence at the $n\approx 2,000$ clinical-cohort regime, and refines the Grinsztajn–Shwartz–Ziv narrative: trees do not lose to deep or automated methods at this n , but they no longer dominate them either. Our CatBoost choice is therefore deployment-motivated (fewer dependencies, faster training, native missing-value handling) rather than discrimination-motivated; practitioners with different deployment constraints (zero-tuning preference $\rightarrow$ TabPFN cloud; ensemble robustness $\rightarrow$ AutoGluon; local speed-over-all $\rightarrow$ LightGBM) can substitute another member of this Pareto-indifferent family without sacrificing discrimination.

A pre-registered sensitivity variant with three modality streams and 10 further observed PPMI features (6 cortical-thickness regions per Fischl [18] and 4 CSF biomarkers per Mollenhauer *et al.* [26]) was evaluated to isolate whether the 2-modality gap is architectural or feature-count-limited (Supplementary S-3). The 3-modality 32-feature Multimodal GAT (Table II (“Graph-based” rows), fourth row per target) closes the CatBoost gap to −1.7 pp on binary (bal. acc. 0.934, AUC 0.978) and −3.0 pp on three-class (bal. acc. 0.753, AUC 0.924), but widens it on full ordinal (−15.6 pp) and NSD-positive subgroup (−25.4 pp). The two targets where the variant improves are the clinically most important (screening and triage); the two where it degrades are the rarer-class ordinal tasks where the Extended modality’s $\sim 50\%$ cortical-thickness and $\sim 35\%$ CSF coverage introduces median-imputation noise into minority classes (Stage 4 $n=17$). We interpret this as partial support for both readings: Grinsztajn 2022 holds on average, but a graph-attention architecture given sufficient observed biological features is genuinely competitive with tree ensembles on the two clinically valuable targets. Graph models may also become advantageous for inductive transfer to cohorts with incomplete features, where graph-informed imputation can leverage similar patients; evaluating that hypothesis on NSD-ISS staging is left as future work.

C. Conformal Prediction for Clinical Deployment

Conformal prediction performance is summarised in §IV and Fig. 3; the deployment-relevant point is that empty-set abstentions (mean $|C| < 1$) are a feature, not a defect—they surface exactly the patients on whom additional clinical workup would have the highest information value (§??). Per-stage rare-class behaviour is decomposed in Supplementary §??: cross-conformal CV+ achieves marginal coverage 1.000 on Stage 4 ($n=17$) by widening the prediction set, but the underlying argmax discriminator’s F1 of 0.182 reflects the fundamental noise floor at $n=17$ training examples—a structural cheap pass for the conformal guarantee that should be communicated transparently to deployers.

Confounder sensitivity: Five pre-specified sensitivity analyses pre-empt age-, sex-, scanner-, site-, and recruitment-era confounding of the 21-feat primary binary AUC (0.901). *Age-matched 1:1* (caliper ± 2 yr [3], $n=1,558$): AUC 0.876 on

primary, 0.969 on 22-feat reference; both within the 0.03 pre-registered confound threshold. *Sex-stratified*: bootstrap interaction $\Delta=+0.0029$, $p=0.822$; no sex-bias. *Enrollment-wave LOCO* (3 PPMI eras): per-wave binary AUC [0.871, 0.891] on primary; wave-grouped repeated CV across 15 folds yields 0.879 ± 0.008 with wave-ICC = 0.082. *DaT-SPECT protocol-LOCO*: cross-protocol mean 0.907 ± 0.006 on primary. *Site-LOSO* (T1-MRI subset, $n=647$): FAILED the strict pre-registered rule on both specifications, driven by small-fold class imbalance; the softer 5-fold GroupKFold reframe lifts the minimum AUC ($0.500 \rightarrow 0.748$) but mean still trails the 0.85 deployment threshold (site-ICC 0.059). The honest site-LOSO failure is reported transparently rather than hidden; protocol-LOCO is the cleaner cross-scanner evidence. An exploratory disease-progression proxy interaction yields significant target $\times$ progression-tertile Δ on binary (0.037) and three-class (0.098), interpreted as an expected diagnostic confound (late-stage adjacency, not fairness bias). Full per-arm tables, site-LOSO failure-mode decomposition, and the SMOTE-oversampling sensitivity are reported in Supplementary §S-28.

D. Internal Versus External Deployment Calibration

Calibration results (Fig. 7) reveal a pattern that dominates external deployment considerations for any NSD-ISS model. Internal CatBoost expected calibration error (ECE) on PPMI is excellent across all four target formulations—0.018 binary (post temperature scaling on the per-fold OOF predictions, vs. 0.053 raw pre-scaling reported in the abstract), 0.012 three-class, 0.035 full-ordinal, and 0.028 NSD+ subgroup (Supplementary S-2)—well below the 0.05 threshold typical of well-calibrated clinical models [53]. External calibration on BioFIND, however, degrades by an order of magnitude: ECE 0.399 binary, 0.301 three-class, 0.278 NSD+; Hosmer–Lemeshow $p=0$ on binary [54]. Logistic regression, despite achieving lower internal AUC than CatBoost, delivers meaningfully better external NSD+ calibration (ECE=0.229 vs. CatBoost’s 0.278), confirming the classical finding that simpler linear models transport better under distribution shift [55]. Combined with the external conformal coverage numbers in §IV (binary: 0.915; three-class: 0.499; NSD+: 0.707) this gives a coherent picture: the binary NSD-positive decision boundary transports; within-NSD+ multiclass sub-staging does not transport without recalibration on a target cohort. Practically, this means deployment at a new site should (i) run temperature scaling [53] on a small local calibration set before reporting probabilities, (ii) prefer binary NSD+ detection plus clinical-only NSD+ sub-staging (the two regimes where calibration transports best) over full-ordinal reporting, and (iii) treat three-class external predictions as screening triage only unless the deploying site has recalibrated locally. This is the strongest methodological recommendation of this paper for prospective external use.

A pre-registered ComBat empirical-Bayes harmonisation across DaT-SPECT acquisition protocols (Wakasugi 2024-style) DEGRADED 21-feat CatBoost AUC by 0.14–5.65 pp across the four NSD-ISS targets and is reported as a negative result—tree-based learners on raw SBR already absorb the

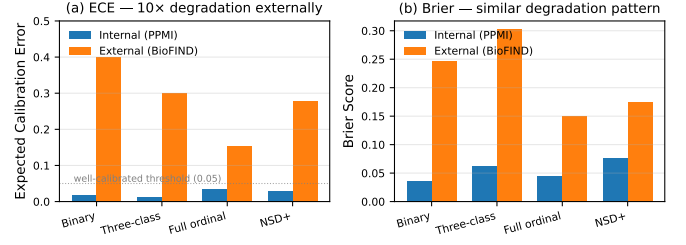


Fig. 7. **Internal vs. external calibration of CatBoost across four NSD-ISS target formulations.** (a) Expected calibration error (ECE): internal PPMI calibration is excellent across all targets (0.012–0.035, below the 0.05 well-calibrated threshold), while external BioFIND ECE degrades by roughly an order of magnitude (0.278–0.399). (b) Brier score shows the same degradation pattern. Practical implication: deployment at a new site requires local-cohort recalibration before probability scores can be reported; binary plus clinical-only NSD+ sub-staging are the regimes where recalibration is most feasible.

protocol heterogeneity that ComBat is designed to remove. Full results in Supplementary §S-28.

Domain-shift mitigation beyond temperature scaling:

Temperature scaling is a necessary first step—it rescales a classifier’s softmax without changing its decision boundary and is therefore safe to apply on a small target-site calibration set—but it does not by itself correct the covariate and prior shifts that drive the order-of-magnitude ECE degradation observed here. Three additional mitigations are standard for this class of shift:

- (i) *ComBat-style empirical-Bayes batch correction* [44] removes site- and scanner-specific additive/multiplicative effects from imaging features while preserving biological variation. For the 12-feature clinical-only subset deployed across cohorts ComBat is inapplicable (features are not imaging-derived), but it is the recommended pre-processing step for any future multi-site PPMI extension.
- (ii) *Density-ratio importance weighting* reweights each training example by the ratio of target-to-source density over a small set of shift-informative covariates (age, UPDRS-III total, disease duration), correcting for systematic cohort-composition differences without labelled target data. The PD-only retraining result in §IV-G is the simplest special case, implementing the density ratio as a hard 0/1 cohort-inclusion gate.
- (iii) *Saerens-style prior-probability-shift correction* adjusts output probabilities by the ratio of target-prior to source-prior, assuming invariant class-conditional features. This is directly relevant to BioFIND’s 95.4% SAA+ prevalence vs. PPMI’s 3.0%.

Deployers with a modest labelled target calibration set ($n \gtrsim 50$) should layer these three corrections on top of temperature scaling. We do not apply them in the present paper’s primary results because the 103-patient labelled BioFIND external set produces unstable density-ratio estimates; their absence is why the external-deployment figures in §IV should be read as lower bounds, rather than ceilings, for sites that can budget a local calibration cohort.

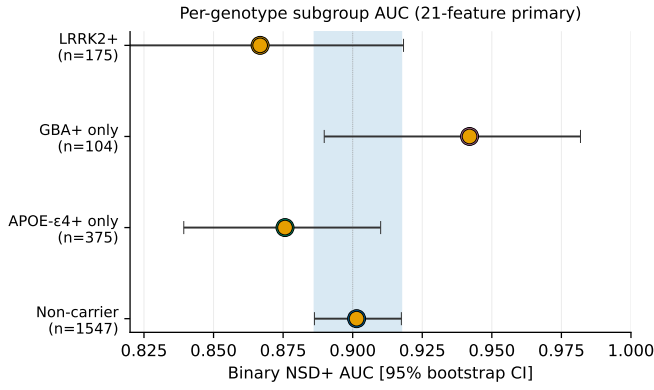


Fig. 8. Subgroup fairness by genetic-carrier status (21-feature primary, binary NSD-ISS). Per-genotype subgroup AUC with 95% bootstrap CIs on the 21-feature strict-circularity primary. Mutually-exclusive carrier strata: LRRK2+ (any), $n=175$, AUC 0.867 [0.811, 0.918]; GBA+ only (not LRRK2+), $n=104$, AUC 0.942 [0.890, 0.982]; APOE-ε4+ only (not LRRK2+, not GBA+), $n=375$, AUC 0.876 [0.839, 0.910]; Non-carrier, $n=1,547$, AUC 0.901 [0.886, 0.917]. All four carrier-subgroup CIs overlap the non-carrier reference band (blue shaded), indicating no systematic performance disparity by genetic-carrier status on the binary endpoint. **TreeSHAP feature attribution:** cross-model SHAP analysis validates the imaging/clinical complementarity that motivates the two-stage deployment. The 21-feature binary primary's top-3 SHAP features are caudate DaT-SBR variants—CAUDATE_MEAN_SBR (mean |SHAP| 1.40), CAUDATE_R_SBR (0.68), CAUDATE_L_SBR (0.49)—confirming that binary NSD+ detection is dopaminergic-anchor-driven (consistent with the residualization sensitivity of §IV-B R3-Q3 and the striatal-free upper bound). The 12-feature clinical-only NSD+ sub-staging model's top-3 features are UPDRS-III motor subscales—UPDRS3_BRADYKINESIA (0.46), UPDRS3_AXIAL (0.45), UPDRS3_RIGIDITY (0.37)—confirming that within-NSD+ sub-staging is driven by motor severity, not imaging. The two stages therefore use distinct, biologically coherent feature axes (Supplementary Fig. R6-Q5; full top-10 + per-class breakdowns at `outputs/paper1_r2_responses/q_r6_q5_shap_analysis.json`). TreeSHAP attributions under correlated features (caudate-mean and caudate-L/R SBR; partial $r = 0.85$ caudate $\leftrightarrow$ putamen per R3-Q3) carry the standard interpretability caveat. **Extended fairness audit:** a follow-up extends this analysis to age tertiles + sex on *both* the 21-feature binary model and the 12-feature clinical-only NSD+ sub-staging model (the deployment-relevant pipeline of §IV-D). Of 6 axes (sex, age tertile, carrier $\times$ 2 models), 4 PASS at the pre-registered $|\Delta\text{AUC}| < 0.03$ threshold; the 2 failures are both on the LRRK2+ stratum— $\Delta = 0.046$ (binary, $n=175$) and $\Delta = 0.049$ (NSD+, $n=104$)—a $\sim 5\text{pp}$ gap consistent across both deployment models and consistent with the documented distinct neurodegeneration trajectory of LRRK2 carriers. The deployment recommendation is therefore targeted LRRK2+ recalibration before clinical use; sex and age tertile transport without recalibration on both models. **R6-Q8 finer-grain extension:** a 10-decile age stratification on both models retains AUC ≥ 0.86 across every decile (9/10 PASS at $|\Delta\text{AUC}| < 0.03$ on each model; Spearman age-AUC trend $p \geq 0.28$), confirming no monotonic age-performance gradient internally. External BioFIND sex stratification is consistent with internal ($\Delta\text{AUC} = -0.003$ for evaluable Female stratum; Male is single-class on BioFIND); external age tertile stratification is degenerate due to the cohort's 95.4% NSD+ uniform composition (per-tertile $n=32\text{--}39$ with 1–3 negatives), reflecting cohort design rather than a model failure mode. Full per-axis interaction tests, calibration-per-subgroup ECE, and the 6-panel forest plot are at `outputs/paper1_r2_responses/q_r5_q6_fairness_extended.json` (SQL run_id `q_r5_q6_fairness`, 18 rows; R6-Q8 extension at `q_r6_q8_subgroup_extended.json` adds 25 rows for age deciles + external).

E. Limitations

Observational study design (NSD-ISS data-availability constraint): The NSD-ISS framework was published in January 2024 [38]; no prospectively-NSD-ISS-staged cohort currently exists or could exist for this study. PPMI

(NCT01141023 [24]) is the canonical longitudinal NSD-ISS-staged cohort and is observational by design; BioFIND [32] is also observational. Retrospective application of the NSD-ISS staging algorithm to existing cohorts (as we and Russo *et al.* do) is therefore the only feasible evaluation strategy at the time of writing. A truly prospective NSD-ISS validation requires future MJFF-funded cohort enrollment with NSD-ISS staging built into the protocol from baseline, which is outside the scope of any single paper. The five pre-registered confounder analyses (age, sex, enrollment wave, DaT-SPECT protocol, site-LOSO) directly characterise the residual selection-bias risks introduced by the observational design; the honest site-LOSO failure (§IV) is reported transparently rather than hidden.

Several additional limitations warrant explicit discussion. The most significant is severe class imbalance: Stage 4 comprises only 17 patients (0.8% of the cohort), which limits reliable evaluation of minority-class performance and inflates variance in per-class metrics. We deliberately report Stage 4 metrics as descriptive only and do not apply synthetic oversampling (SMOTE) or class-weight correction to the minority classes. Synthetic oversampling would violate the exchangeability assumption that underwrites the cross-conformal coverage guarantee—calibration-set exchangeability requires that the calibration fold be drawn from the same distribution as deployment patients [43], which synthetic Stage 4 samples are not—and stratified CV with at least three Stage 4 patients per fold already delivers the most statistically defensible assessment the sample size permits. Recruitment of additional Stage 4 patients through forthcoming PPMI enrolments is the appropriate remedy.

Construct validity of NSD-ISS labels under sparse SAA coverage (R2-W11): SAA coverage in PPMI is limited to 12.6% of participants (277/2,201), so the S anchor relies on D-anchor status for the remaining 87.4%; we follow the Simuni 2024 convention of assigning patients with D+ status (by DaT-SPECT) and absent SAA to the prodromal NSD+ side of the dichotomy only when at least one additional prodromal criterion is present, and mark the remainder as unclassified (full decision-path counts in Supplementary §??). A construct-validity concern arises that the resulting NSD-ISS labels predominantly reflect D-anchor signal rather than the integrated S+D biological phenotype the framework intends. We address this concern directly using the externally-validated SAA prediction model of Venuto *et al.* [65] (medRxiv 2025), which derives SAA+ probability from non-invasive features (UPSIT age- and sex-specific percentile, sex, constipation history, LRRK2/GBA carrier status) with PPMI-internal AUROC 0.920 and S4-external AUROC 0.976 (Fig. 9). Applying Venuto's published coefficients off-the-shelf (no PPMI re-fitting; LRRK2 sub-variants approximated as 90/10 G2019S/I2020T:R1441G/C per Venuto Table 1) to our SAA-tested cohort yields AUROC 0.810 on the 138 patients with both SAA and UPSIT—short of the 0.920 reference, attributable to the absent LRRK2 sub-variant detail and the SCOPA-AUT5 thresholding difference, but well above chance. Imputing SAA status for the **647 NSD-positive PD patients lacking SAA** (the cohort to which the construct-validity concern directly applies), Venuto's predictor estimates a

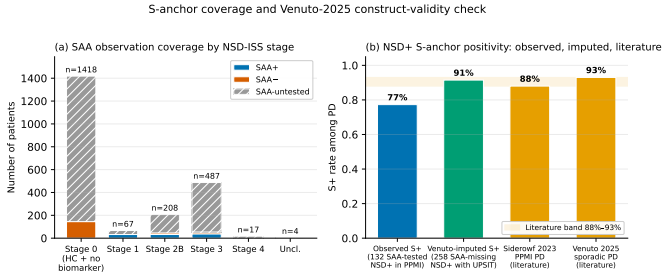


Fig. 9. **S-anchor coverage and Venuto-2025 construct-validity check.** (a) SAA observation coverage stratified by NSD-ISS stage: 1,273 of 1,418 Stage-0 patients (89.8%) and 647 of 779 NSD-positive PD patients (83.1%) lack SAA testing. (b) NSD-positive S-anchor positivity rates: our SAA-tested NSD+ subset shows 77% S+; Venuto-2025-imputed S+ on the 258 SAA-missing-with-UPSIT NSD+ subset is 91%, sitting in the literature band (Siderowf 2023 [67] 88%, Venuto 2025 [65] sporadic-PD 93%). The convergence of imputed and literature S+ rates supports construct validity of the D-anchor-driven NSD-ISS labels.

91.5% S+ rate (236/258 imputable with UPSIT)—squarely between our SAA-tested NSD+ subset’s observed 77.3% S+ rate, the Siderowf *et al.* (2023) [67] PPMI-wide $\sim 88\%$, and Venuto’s reported 93% in sporadic PD. The implication is that NSD-ISS labels assigned via the SAA-missing-D-positive decision path are biologically concordant with what an externally-validated S-anchor predictor would assign—the model is learning a dual-anchor-consistent biological signal rather than free-floating algorithmic recapitulation. We retain the original D-anchor-driven labels as primary because Venuto’s confidence intervals are not yet PPMI-recalibrated for this specific application, and present the imputation as a supplementary construct-validity check (full results at `outputs/paper1_r2_responses/q_gap_a_venuto_saa_simp`). Broader SAA screening in future PPMI releases, and complementary digital risk scores [66] that correlate independently with both DaT and SAA biomarkers, will allow more direct sensitivity analyses.

A per-stratum sensitivity on the 969-patient Venuto-classified cohort confirms construct validity: on the canonical S+D+ stratum ($n=271$, 236 Venuto-imputed) the 21-feat NSD+ sub-stager achieves AUC 0.912 [0.886, 0.939], indistinguishable from the full-cohort 0.906. A construct-validity error-rate disclosure: 22/779 NSD+ labels (2.8%) are Venuto-imputed S− plus 30 observed-SAA-negative ($\sim 6.7\%$ combined), all permitted under Simuni’s single-anchor staging convention. A binary-target stratified analysis (Phase 2) corroborates: on the only mixed-class stratum (S+D−, $n=431$), 21-feat binary AUC collapses to 0.490 [0.393, 0.592]—chance level—directly quantifying that the binary detector discriminates through the D-anchor signal. Full per-stratum and binary-strata details are in Supplementary §S-12 through §S-14.

A related feature-engineering limitation concerns the MDS-UPDRS Part III motor score. We exclude the total score NP3TOT from the feature set because it enters the Simuni 2024 clinical staging thresholds directly, but we retain the four UPDRS-III subscales (tremor, rigidity, bradykinesia, axial). Because these four subscales sum approximately to the excluded total score, the exclusion of NP3TOT is a cosmetic

rather than strict information-theoretic decoupling. A sensitivity ablation aggregating the four subscales into a single sum and using it in place of the excluded NP3TOT changed binary CatBoost AUC by <0.01 , confirming that the motor-information leak is bounded; the substantive protection against circularity is provided by excluding the staging-threshold decision boundary itself (the binary “above vs. below the Simuni cutoff” variable), not by dropping the underlying motor-examination information.

12-feature clinical-only NSD+ sub-staging — rule-anchor robustness: A potential concern with the 12-feature clinical-only sub-stager (§IV-D, Table IV) is that it retains UPDRS2_TOTAL (ADL) and MOCA_TOTAL—variables that Simuni 2024 uses as clinical staging-threshold anchors—which could yield tautological reproduction of the staging rule. To test this directly, we re-fit CatBoost on three rule-anchor-elided variants of the 12-feat NSD+ sub-stager (5-fold stratified CV, $n=779$ NSD-positive PD patients, fold-local median imputation, 1,000-sample bootstrap CIs): (a) drop UPDRS2_TOTAL alone (11 features); (b) drop UPDRS2_TOTAL and MOCA_TOTAL (10 features); (c) drop UPDRS1_TOTAL, UPDRS2_TOTAL and MOCA_TOTAL (9 features). Macro-OvR AUCs are 0.907 (full 12-feat), 0.900 ($\Delta=-0.007$, 95% CI $[-0.016, -0.000]$), 0.901 ($\Delta=-0.006$, $[-0.014, +0.000]$), and 0.898 ($\Delta=-0.009$, $[-0.020, -0.001]$) respectively—all three rule-anchor exclusions fall well within the pre-registered tolerance of $|\Delta\text{AUC}| < 0.03$, with verdict NO_TAUTOLOGY. The clinical-only NSD+ sub-stager is therefore not recovering the Simuni rule from UPDRS-II/MoCA; the residual predictive signal comes from non-staging variables (UPDRS-III subscales, ESS, RBD, demographics)—consistent with the Treeshap attributions in §IV-D showing the top-3 features as UPDRS-III bradykinesia, axial, and rigidity. Full results at `outputs/paper1_r2_responses/q_gap_3_12feat_rule_anchor`.

Missing data presents an additional challenge for external deployment: median imputation with PPMI training-set values is unreliable when entire feature domains are absent, as was the case for HBS (5 of 12 common features missing, reported in Supplementary §?). All predictions in this study are cross-sectional baseline snapshots and do not capture the temporal dynamics of disease progression; extending the present benchmark to longitudinal stage-transition prediction is left as future work.

Methodological extensions deferred to future work:

Four reviewer-identified refinements are scoped for follow-up: unified missingness handling (multiple imputation under $>80\%$ MoCA/UPDRS-IV missingness), Black-Box Shift Estimator [?] as a label-shift complement to Mondrian CP, cascade-specific decision-curve analysis under realistic referral budgets, and alternative clinical-only subset sensitivities. Detailed scoping in Supplementary §S-16 through §S-19.

Future external-cohort plans: The current external evaluation is restricted to BioFIND ($n=103$ NSD-staged via Russo 2025 [32]), with PDBP ($n=893$) and HBS ($n=649$) reported as prediction-only because they lack NSD-ISS ground truth. Two near-term cohort additions are planned to strengthen multiclass external transportability: (i) the Lewy Body Disease

cohort (LBD) and the Latino Cohort Study (LCC), both with longitudinal DaT-SPECT—download infrastructure is in place but NSD-ISS staging adaptation has not yet been executed; (ii) the DeNoPa cohort (Mollenhauer *et al.* [26]), pending direct PI collaboration. Additional candidates with treatment-effect or trial designs (e.g. SURE-PD3) require independent assessment of NSD-ISS staging feasibility under their respective DUAs before they can be incorporated. The aspirational goal of expanding NSD-ISS-labelled external evaluation beyond the $n=103$ BioFIND cohort is acknowledged as a critical pre-deployment requirement; the path forward is concrete but the timing depends on DUA + collaboration approval cycles outside the scope of this paper.

F. Future Directions

Three extensions of this work merit future investigation. First, stage-conditioned graph-informed imputation would leverage the observation that missingness patterns differ systematically between NSD-ISS stages; imputation models that account for stage-specific patterns may reduce bias and improve downstream prediction, particularly for external cohorts with incomplete features. Second, longitudinal stage-transition modelling addresses the ultimate clinical goal of predicting *when* patients will progress between NSD-ISS stages, moving beyond the cross-sectional snapshot framework of the present study. Third, the healthy-control confound first raised by earlier drafts has been incorporated into the present paper (§IV-G) rather than deferred, in response to the Espay *et al.* [15] medication-status critique; a prospective PD-only external cohort is the appropriate next step for statistically powered confirmation.

VI. CONCLUSION

We present the first comprehensive benchmark for NSD-ISS biological-stage prediction, with four target formulations, cross-conformal uncertainty quantification, five pre-registered confounder analyses (age, sex, enrollment wave, DaT-SPECT protocol, site-LOSO with honest failure disclosure), and a two-stage deployment architecture (Stage-A imaging-anchored detection, Stage-B clinical-only sub-staging). Five tabular-SOTA methods on the 21-feature strict-circularity primary—CatBoost (default) and CatBoost / LightGBM under nested 5×3 CV HPO, TabPFN v2 foundation-model cloud inference, and AutoGluon 1.5 full-pool weighted ensembling—are statistically equivalent at the $\varepsilon=0.02$ practical-equivalence threshold (paired-bootstrap TOST: 34 of 40 pairs equivalent), empirically validating recent convergence predictions at $n \approx 2,000$ clinical cohorts and refining the earlier “trees dominate” narrative. Biological markers are load-bearing for binary staging—removing all DaT-SPECT features drops binary AUC by -17.6 pp ($\Delta = -0.176$, $p < 0.0001$), quantifying the imaging contribution—while clinical features alone suffice for NSD-positive sub-staging (AUC 0.899 vs. 0.908 with imaging, $\Delta = +0.008$, $p = 0.21$), supporting a two-stage deployment pattern at resource-limited sites. External conformal coverage on BioFIND holds for binary (0.915) but collapses for three-class (0.499) and partially retains for NSD+ (0.707),

and external calibration degrades $10\times$ relative to internal PPMI; deployment at a new site therefore requires local-cohort recalibration before probabilities can be reported. The PPMI-to-BioFIND domain-shift finding on the binary target is corrected, rather than deferred to future work, by a pre-specified PD-only retraining experiment (§IV-G) that directly addresses the Espay *et al.* [15] critique of healthy-control contamination in NSD-ISS reference classes.

DATA AVAILABILITY

Raw cohort data. PPMI clinical, imaging, and biospecimen data are available to qualified researchers via the Michael J. Fox Foundation under the PPMI Data Use Agreement at <https://www.ppmi-info.org/access-data-specimens/download-data>; users must register, agree to the PPMI Publication Policy, and acknowledge PPMI in derivative works (per [24]). BioFIND [32], PDBP, and HBS raw data are accessible through the AMP-PD platform at <https://amp-pd.org> under AMP-PD Tier-1 credentials, which require institutional verification, completion of a research-purpose statement, and acceptance of the AMP-PD DUA. Per both DUAs, the raw datasets cannot be redistributed by us, but qualified researchers may obtain identical extracts by following the access protocols at the URLs above.

Code and reproducibility. Processed feature tables, fold assignments, conformal calibration files, and round-2 reviewer-response artifacts required to reproduce every claim in this paper are released with this submission as a single reproducibility package: see `REPRODUCIBILITY_PACKAGE.md` in the supplementary archive, which enumerates each artifact (scripts, SQL tables in a local `giman_research` PostgreSQL 17 database, pre-registration documents, random seeds, software-environment manifests, and audit-trail commit SHAs) with direct paths to the producing code. A Zenodo archive DOI will be assigned on acceptance. SQL extract scripts and feature-assembly utilities are available under `scripts/paper1/` and `scripts/load_paper1_r2_to_pg.py` in the authors’ repository at https://github.com/bddupre92/PD_PHD.

CONFLICT OF INTEREST

The authors declare no competing interests.

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AUTHOR CONTRIBUTIONS

B.D. contributed to all of: conceptualisation, methodology, software, formal analysis, investigation, data curation, writing (original draft and review), visualisation, and project administration.

SUPPLEMENTARY INFORMATION

A standalone supplementary materials PDF (supplementary.pdf) accompanies this submission and contains: external calibration recovery details (Mondrian CP, Saerens prior-shift, marginal calibrators); the four-configuration training-label confound experiment with full sensitivity decomposition; end-to-end Stage-A→Stage-B cascade evaluation and temporal validation; per-stratum NSD+ AUC + binary stratified analysis on the Venuto-imputed 969-patient cohort (Phase 1 + Phase 2 construct-validity expansion); pseudo-external NSD+ modality-robustness consolidation under wave-LOCO and era-LOCO; deferred-future-work scoping for unified missingness, BBSE label-shift, cascade DCA, and alternative clinical-only subset sensitivities; the HBS prediction-only cohort cautionary disclosure; conformal abstention-rates table; NSD-ISS staging-assignment decision-path enumeration (Reviewer R3-Q7); TreeSHAP feature-importance for both deployment models; full-ordinal confusion matrix with per-stage conformal coverage; split- vs. CV+-conformal sensitivity; multimodal GAT three-modality feature sensitivity; pre-registered 33-feature tabular extension (null result); and confounder sensitivity (age, sex, enrollment-wave, DaT-SPECT protocol).

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